



BIODES BIOGAS RECOVERY AND ELECTRICITY GENERATION



Document Prepared by



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1 PROJECT DETAILS

1.1 Summary Description of the Project

The BiodeS biogas recovery and electricity generation project (hereafter, the “Project”), developed by Agro de Souza (hereafter, also referred as the “project proponent”), consists in the implementation of an anaerobic manure management system for methane recovery and energy generation.

Agro de Souza is located in Coronel Suarez, Buenos Aires Province, Argentina. The farm has a total area of 99 hectares, where 93 are destined to agriculture and the rest involves a feedlot with an installed capacity of 3,000 cows.

Before the 19th of September 2022, the capacity of the establishment was about 1,000 heads, and the effluents were permeating the soil without any management. However, due to the expansion of the number of animals, new measures had to be implemented. The common practice in Argentinian establishments of this dimension, is to deliver effluents to an open-air anaerobic lagoon. Due to the environmental damage that this technology causes, and the large amount of methane emissions generated, Agro de Souza decided to implement two biodigesters for manure management, to capture methane emissions from the biogas produced, which will also displace fossil fuel energy generation. In case that the gas cannot be used in the cogeneration unit, it will be burned on a flaring torch.

The project consists of three roofed confined areas with concrete floors, connected by load ways which lead to a main loading chamber. The manure is concentrated in this compartment and delivered to the biodigesters. Separately, the vegetable silage is accumulated in the solid’s loader, where it is crushed and mixed with digestate, to be pumped to the primary digesters. The generated biogas in the digesters is stored in the upper part, with a maximum volume of 780 m³. These have a biogas engine-generator set (genset) of 0.635 MW that produces electricity to be dispatched to the Argentinian Interconnection System (hereafter, SADI). The digestate from the biodigesters is used as biofertilizer in the farmlands under aerobic soil applications. In addition, the generator produces thermal energy as hot water that will be used to heat the digesters.

Therefore, emission reductions are achieved by i) recovering the methane from the biogenic organic mixture co-digested in the anaerobic digesters and ii) renewable energy generation.

The estimated annual average of GHG emission reductions is 14,837 tCO₂e/year, reaching a total of 103,052 tCO₂e in the first seven years.

1.2 Sectoral Scope and Project Type

Sectoral Scope:

- Scope 13 – Waste handling and disposal.
- Scope 1 – Energy Industries – renewable/non-renewable sources

The project is a Non-AFOLU project.

The project is not a grouped project.

1.3 Project Eligibility

The scope of the VCS Program includes:

- 1) The six Kyoto Protocol greenhouse gases: The emissions reduction of this project comes from: CH₄ from anaerobic digestion recovered and destroyed to avoid CH₄ released to the atmosphere; and CO₂ emissions from the electricity generation by replacing fossil fuel fired power plants.
- 2) Ozone-depleting substances: This project does not include ozone depleting substances.
- 3) Project activities supported by a methodology approved under the VCS Program through the methodology approval process: The methodology applied in this project is not approved under the VCS program. Through a methodology approval process
- 4) Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval: The methodology AMS-III.D. (Version 21.0) applied by the project is under the CDM Program which is a VCS approved GHG program.
- 5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in VCS document JNR Requirements: Not applicable.

According to section 3.1 of VCS Standard (Version 4.4):

Establishing a consistent and standardized certification process is critical to ensuring the integrity of VCS projects. Accordingly, certain high-level requirements must be met by all projects, as set out below.

Reference to VCS Standard (Version 4.4)	Requirements	Justification
3.1.1	Projects shall meet all applicable rules and requirements set out under the VCS Program, including this document.	The project meets all applicable rules and requirements set out under the VCS Program, including this

	Projects shall be guided by the principles set out in Section 2.2.1.	document.
3.1.2	Projects shall apply methodologies eligible under the VCS Program. Methodologies shall be applied in full, including the full application of any tools or modules referred to by a methodology, noting the exception set out in Section 3.14.1. The list of methodologies and their validity periods is available on the Verra website.	The project applies the methodology AMS-III.D. (version 21.0) along with tools or modules as applicable developed under the United Nations Clean Development Mechanism, which can be used for projects and programs registering with VCS as per requirements available on the Verra website.
3.1.3	Projects shall apply the latest version of the applicable methodology in all cases unless a grace period applies to the project as set out in 3.21 below. Projects must update to the latest version of the methodology when reassessing the baseline and renewing a crediting period.	The project applies the methodology AMS-III.D. to the latest version (version 21.0).
3.1.4	Projects and the implementation of project activities shall not lead to the violation of any applicable law, regardless of whether or not the law is enforced.	The project is in compliance with applicable law, regardless of whether or not the law is enforced. According to the environment impact assessment report (EIA), the project complies with all Argentinian relevant laws and regulations.
3.1.5	Where projects apply methodologies that permit the project proponent its own choice of model (see the VCS Program document Program Definitions for definition of model), such model shall meet with the requirements set out in the VCS Program document VCS Methodology Requirements and it shall be demonstrated at validation that the model is appropriate to the project circumstances (i.e., use of the model will lead to an appropriate quantification of	Not applicable as there is no model that needs to be chosen by the project proponent as per the applied methodology AMS-III.D. (version 21.0).

	GHG emission reductions or removals).	
3.1.6	Where projects apply methodologies that permit the project proponent its own choice of third-party default factor or standard to ascertain GHG emission data and any supporting data for establishing baseline scenarios and demonstrating additionality, such default factor or standard shall meet with the requirements set out in the VCS Program document VCS Methodology Requirements.	Not applicable as there are no third-party default factors or standard need to be chosen by the project proponent as per the applied methodology AMS-III.D. (version 21.0).
3.1.7	Where the rules and requirements under an approved GHG program conflict with the rules and requirements of the VCS Program, the rules and requirements of the VCS Program shall take precedence.	All rules and requirements from the VCS program will be used as a preference than the CDM requirements.
3.1.8	Where projects apply methodologies from approved GHG programs, they shall comply with any specified capacity limits (see the VCS Program document Program Definitions for definition of capacity limit) and any other relevant requirements set out with respect to the application of the methodology and/or tools referenced by the methodology under those programs.	The methodology requires that the measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO ₂ equivalent annually from all Type III components of the project activity. This project does not exceed the limit of 60 kt CO ₂ equivalent annually.
3.1.9	Where Verra issues new VCS Program rules, the effective dates of these requirements are set out in Appendix 3 Document History and Effective Dates or equivalent for other program documents and are listed in a companion Summary of Effective Dates document which corresponds with each update.	For the presentation of this project, it was considered the last version of the “overview of substantive updates to program rules and requirements and effective dates”, released the 21 of December of 2022.

1.4 Project Design

As described in Section 1.1, the project activity includes a single location and has not been developed as a grouped project. Location is described in Section 1.12 and the installed technology is in Section 1.11 of this document.

Eligibility Criteria

The project is not a grouped project; therefore, this section is not applicable.

1.5 Project Proponent

Organisation name	Agro de Souza S.A.
Contact person	Cecilio de Souza
Title	President
Address	Provincial route 85 km 190 – Coronel Suárez, Buenos Aires Providence
Telephone	+54 9 2926 431658 / +54 9 2923 659085
Email	contacto@agrodesouza.com.ar biogas@agrodesouza.com.ar

1.6 Other Entities Involved in the Project

Organisation name	Pike Carbosur SA
Role in the project	Project Developer
Contact person	Álvaro Pérez del Castillo
Title	Executive Director
Address	Costa Rica 1566, CP 11500, Montevideo, Uruguay.
Telephone	+598 2605 6231
Email	alvaro.perez@carbosur.com.uy

1.7 Ownership

The project owner is Agro de Souza S.A. The documents that prove the project ownership will be provided to the VVB as required.

1.8 Project Start Date

The project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions. This date is the 19th of September 2022.

1.9 Project Crediting Period

Crediting period: from 19/09/2022 to 18/09/2029.

7 years twice renewable (21 years).

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

- ☒ <20,000 tCO₂e/year
- ☐ 20,000 – 100,000 tCO₂e/year
- ☐ 100,001 – 1,000,000 tCO₂e/year
- ☐ >1,000,000 tCO₂e/year

Project Scale	
Project	√
Large project	

Year	Estimated GHG emission reductions (tCO ₂ e)
From 19/09/2022 to 31/12/2022	3,346
2023	14,512
2024	14,512
2025	14,512
2026	14,512
2027	14,512
2028	14,512
From 01/01/2029 to 18/09/2029	10,376
Total estimated Ers	100,796
Total number of crediting years	7
Average annual Ers	14,399

1.11 Description of the Project Activity

The project consists in biogas recovery and electricity generation through the implementation of an anaerobic manure management system to recover methane and generate energy.

The farm has 3 barns (subdivided into 8 corrals each) with roof and concrete floor, with a maximum capacity of 3,000 livestock heads. The manure is collected daily by scrapers that sweep it into the loading channels and then into the loading chamber after dilution. Subsequently, two agitators homogenise the mixture, which is fed to the primary digester by a vertical pump.

Both biodigesters are equipped with agitators and an internal heating system, where anaerobic conditions are maintained at 37° C. The plant will have a production capacity of 2,163,936 Nm³/year considering a continuous production flow of 247 Nm³/h, with around 54,8% of methane.

Once the biogas is generated, it is purified and dried to enter the internal combustion engines for electric power generation. Since the biogas flow is continuous, the energy production is constant and is connected to the SADI. If it cannot be used in the cogeneration unit due to maintenance work, it will be burned in a flaring torch with a capacity of 300 Nm³/h. The thermal energy produced in the generator will be used to heat the biodigesters as hot water.

The digestate obtained from the fermentative process is divided into biol (liquid) and biosol (solid) by a solid separator. Part of the biol goes to a lagoon where it is stored for 0 to 60 days to be used as biofertilizer and part is redirected to the biodigester. Biosol is composted aerobically and then spread on the farm as biofertilizer. Annual production of digestate is estimated at 40,000 tons of fresh material.

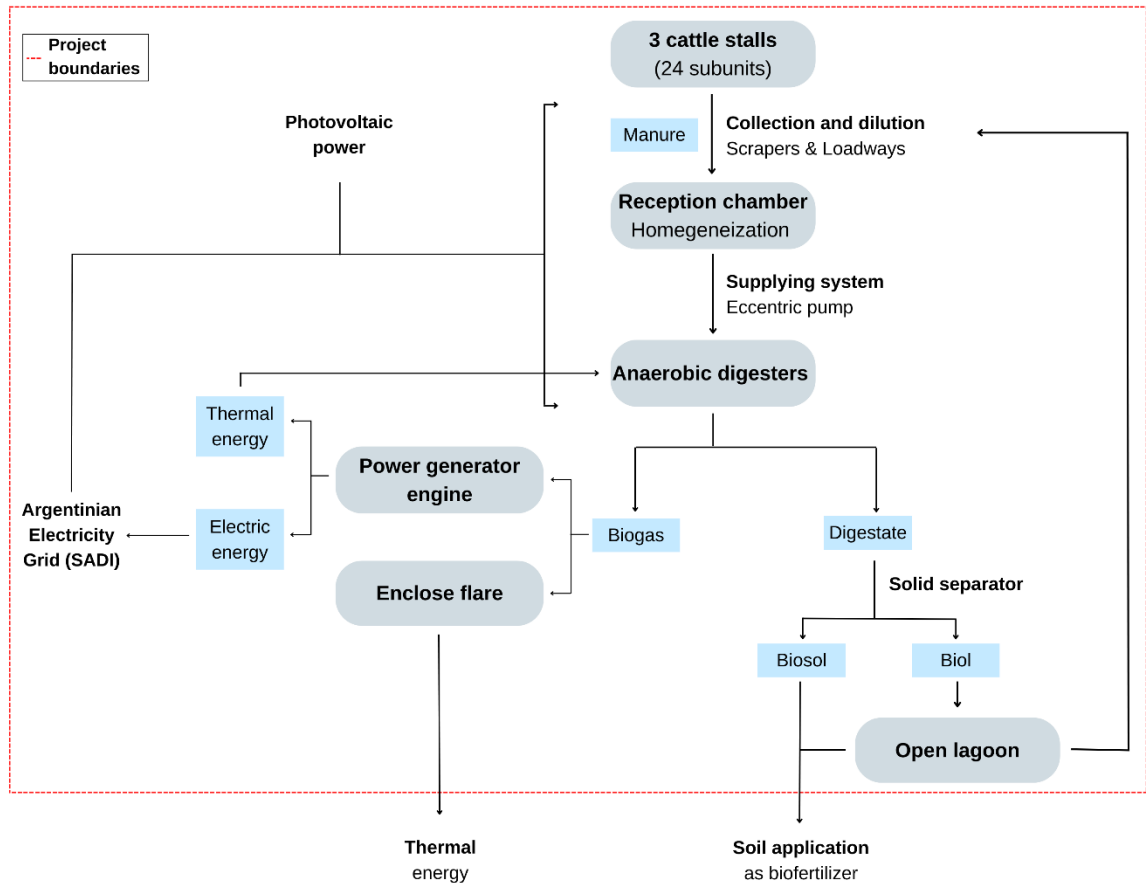
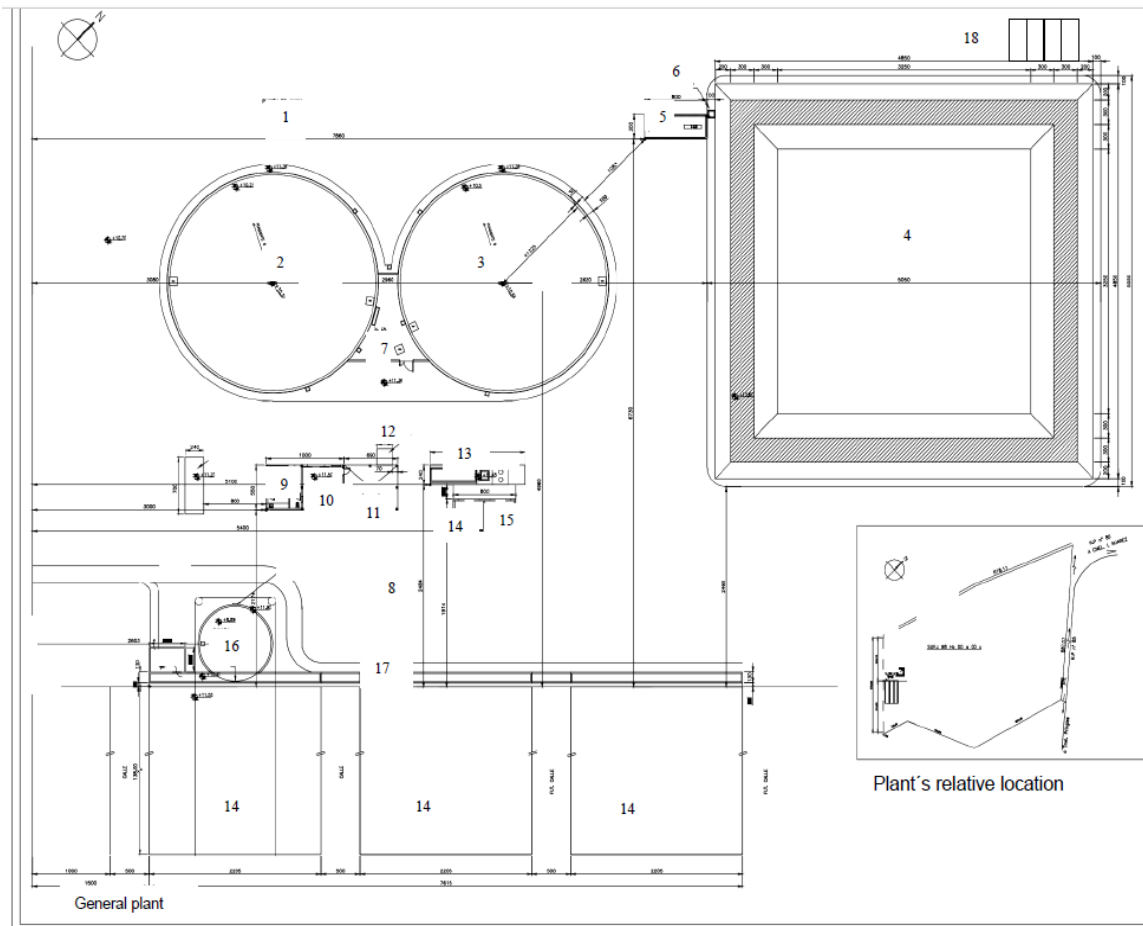


Figure 1. Process map of BiodeS



- | | | | |
|--------------------|------------------------|------------------------------------|---------------------|
| 1. Loading dock | 6. Leachate chamber | 11. Semi-covered | 16. Loading chamber |
| 2. Biodigester 1 | 7. Bomb | 12. Uncovered plate for the filter | 17. Loading channel |
| 3. Biodigester 2 | 8. Enclose flare | 13. Power generator engine | 18. Cells |
| 4. Lagoon | 9. Control room | 14. Corrals | |
| 5. Solid separator | 10. Auxiliary services | 15. Cell room | |

Figure 2. Facilities of BiodeS project

Table 1. Equipment description of BiodeS plant

Equipment	Biodigesters	Flaring torch	Electricity generator
Number of equipment	2	1	1
Type	Circular overhead	Thermostatically controlled jacket – FAII300	JGS 312 GS-S,L Serial N° 1104040
Material	Concrete H30	AISI 316 & AISI 304	Steel
Fabrication date	2022	2021	2014
Capacity	3,034 m ³	0.194 MWh	0.635 MWh
Dimensions	Diameter: 26.5 m Hight: 7.5 m	1 m ²	Length: 12.2 m Width: 3.6 m Hight: 6.7 m
Lifetime (years)	20 years*	20 years*	20 years*
Load factor	N/A	N/A	78%
Efficiency		300 Nm ³ /h	42%

*The equipment is designed for an operational lifetime according to the project lifetime (20 years). The equipment is periodically maintained to ensure the correct functioning of all the parts.

1.12 Project Location

The project is located in Coronel Suárez, km 166 of route 85, at Buenos Aires Providence. The geographic coordinates are south latitude 37°39'21.18'' and west longitude 61°52'26.62''.



Figure 3. Project location

1.13 Conditions Prior to Project Initiation

In 2001 the company started with the development of infrastructure of the first feedlot, with a capacity of 1,000 heads. These were the conditions prior to the project implementation, where effluents had no treatment associated and were collected every 3 days and stacked in a pile.

Once the company proposes to expand its livestock capacity, it becomes impossible to continue with this management from an operational and environmental point of view. Therefore, Agro de Souza started discussing the possible alternatives for this productive dimension.

The conditions existing prior to the project initiation are not the same as the baseline scenario. The last one is described in Section 3.4

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The local, regional, and national laws, statutes and regulatory frameworks, that are required for this type of projects in Argentina are:

National level:

- National Constitution
- Law N° 24,065 “Electric Energy Regime” and its Decree N°1,398/92
- Law N° 25,675 “General Environment Law”
- Law N° 26,190 “Renewable Sources of Energy Promotion Regime”
- Law N° 24,051, Decree N° 831/93 “Dangerous residues”
- Resolution N° 70/2001 and 329-E/2017 of the “National Registry of Feedlot Livestock Establishments”.
- Resolution N° 182/95
- Authorization to be a Wholesale Electricity Market (MEM, due to its name in Spanish) agent to the Argentinean Ministry of Energy.

Buenos Aires Province:

- Buenos Aires Province Constitution.
- Ordinance N° 6957/18 of Coronel Suarez.
- Provincial Law N° 11.723 of “Protection, conservation, improvement and restoration of natural resources and the environment in general”.
- Provincial Law N° 5965 “for the protection of water supply sources, water courses and receiving water bodies, and the atmosphere.

The Project complies with all national law and regulations since the Buenos Aires Province Government approved the Resolutions N° “IF-2022-31332861-GDEBA-DPEIAMAMGP” the 19th of September of 2022 and N° “RESO-2022-565-GDEBA-SSCYFAMAMGP” the 2nd of December of 2022.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has not been registered nor is seeking registration under any other GHG programs. It will only be registered in the VCS program.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected nor presented in any other GHG program.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project proponent is not part of any emission trading program nor any other mechanism that includes GHG allowance trading and will not use the project's GHG emission reductions for compliance with these programs, nor mechanisms. Furthermore, Argentina does not have a GHG emission compliance nor trading programs.

1.16.2 Other Forms of Environmental Credit

The project proponent has not sought nor received another form of GHG-related credit, including renewable energy certificates. The project has not sought nor received another form of GHG-related environmental credit, that is why, the emission reductions generated of this project, will not be double counted.

Supply Chain (Scope 3) Emissions

Supply Chain (Scope 3) emissions are assessed in line with the VCS Standard 4.4 and Corporate Value Chain (Scope 3) Accounting and Reporting Standard. According to these sources, this section is not applicable since the project does not directly impact emissions associated with any good or service other than the electricity supplied to the national grid. However, according to the statement given at the Corporate Value Chain (Scope 3) Accounting and Reporting Standard "The generator does not account for scope 3 emissions associated with sold electricity because the emissions are already accounted for in scope 1." Consequently, the project activity is not related with any scope 3 emissions.

1.17 Sustainable Development Contributions

Argentina is one of the 150 countries that adopted the 2030 Agenda for Sustainable Development¹, including the Sustainable Development Goals (hereafter, SDG²) of the United Nations. The project contributes with at least 3 SDG listed above:

- SDG 7: Affordable and Clean Energy.

BiodeS project contributes with SDG 7 by displacing the use of fossil fuels in electricity generation with biogas produced in biodigesters. This renewable energy is supplied to

¹ Argentina's objectives, goals and indicators for sustainable impact

https://www.argentina.gob.ar/sites/default/files/cncps_-_agenda_2030_objetivos_de_desarrollo_sostenible_junio_2021_v7_2.pdf

² Sustainable Development Goals - SDG <https://www.un.org/sustainabledevelopment/sustainable-development-goals/>

SADI to contribute to the access of affordable, reliable, and modern energy services. It is estimated that the project will have a net production of 4,641 MWh per year.

Argentina will measure the compliance of this goal through the percentage of renewable energy and the percentage of the population with access to electricity.

- SDG 8: Decent Work and Economic Growth

Not only the construction and installation of equipment provided new employment opportunities, but also other companies participated in the development of the technologies and the design of the project. This probably generated a contribution in the regional and national economy. Therefore, during the project, direct and indirect employment will be requested, which will increase the local economy through the hiring of residents.

The national government will measure this impact through the growth of the Gross Domestic Product (GDP) and the employment rate.

- SDG 9: Industry, Innovation and Infrastructure

BiodeS has upgraded its infrastructure to make it sustainable, increasing its resource-use efficiency and adopting a cleaner and environmentally sound technology. This project also enhances scientific research and upgrades the technological capabilities of the sector, encouraging innovation and increasing the number of research and development workers.

The national goal is to increase the Investment in research and development activities and measure it as a percentage of GDP and measure the greenhouse gas emissions in reference to it.

- SDG 13: Climate Action

The main impact of this project is the avoidance of methane emissions that otherwise would be liberated to the atmosphere, producing a negative impact on climate change. In addition, CO₂ emissions are reduced due to the use of renewable sources for energy generation, instead of fossil fuels. The expected emissions reduction during the crediting period (19th of September of 2022 to the 18th of September of 2029) are 103,057.05 tCO₂e, with an annual average of 14,722.44 tCO₂e. From these the 90% correspond to avoided emissions from the recovery of methane through biogas production, while only 10% correspond to the generation of renewable energy.

Argentina intends to promote effective planning and management in relation to climate change, measuring total greenhouse gas emissions of the country.

The National Council for the Coordination of Social Policies (CNCPS, due to its name in Spanish) is the entity responsible for the development and monitoring of these goals.

1.18 Additional Information Relevant to the Project

Leakage Management

According to the methodology used (AMS III.D. version 21) leakage effects are considered only if the equipment used in this project is transferred from another activity or if the existing one prior to project implementation is transferred to another. The present technology consists of new equipment, not transferred from another project, nor will it be transferred to another activity, so it is not needed to specify a leakage management plan nor the leakage and risk mitigation measures.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

No further relevant information is identified.

2 SAFEGUARDS

2.1 No Net Harm

The EIA of the project was completed in December 2022 and approved by the Buenos Aires Province Government. This report declares that the negative impacts that the project can have in reference to the atmospheric contaminants, are not significant nor superior to the maximum limits defined in the air quality resolutions established by the Provincial Organism for the Sustainable Development of Buenos Aires Providence, Law N° 5,965 and the Decree N° 1074/2018. Additionally, the EIA defines that the project is under the limits established for atmospheric emissions by the Decree N° 831/93 and the Resolution 182/95 due to the fact BiodeS is under the national jurisdiction by being a MEM agent.

Is important to mention that the insignificant negative impacts that the project may have, are irrelevant in comparison to all the positive impacts described in the SDGs, such as: generating electricity from renewable sources, reducing greenhouse gas emissions, obtaining a biofertilizer rich in nutrients and generating jobs.

2.2 Local Stakeholder Consultation

On the 2nd of March of 2023 the biogas plant was formally presented to the authorities and the local stakeholders. The meeting was held in BiodeS, in the project location.

The invitations were sent via email and between the guests were the authorities listed below:

- Ricardo Alejo Moccero (Mayor of Coronel Suarez)
- Florencia Terán (Director of Renewable Energies of the National Energy Secretariat)
- Carolina Digiani (Chief of Staff of the Undersecretariat of Environmental Policy of the Ministry of Environment of the Province)

The project was presented by its owners and the authorities mentioned the importance of BiodeS for the environment of Coronel Suarez and Argentina.

Table 2. BiodeS local presentation

Agenda	
11:00	Opening of the meeting
11:30	Introduction to the project, positive environmental impacts, and energy generation.
12:30	Ribbon cutting ceremony
13:00	Tour with authorities and local stakeholders in the plant.
13:30	Lunch

The local stakeholders have an email address, and a phone number were the questions and doubts are answered. There were no questions about the project, only positive comments, and requests to visit the biogas plant. The email and the phone number are still active for receiving any comment or questions.

Before this meeting, the project has been presented to the society in many opportunities since local and national media have visited BiodeS. The project also received the national approval in 2022. It was also presented as a case of study “*Adaptation of production systems to sustainability challenges*” CHIMENO, Patricia; POLCARO, Sofía; SALDUNGARAY, María Cecilia; TORRES CARBONELL, Carlos; PIÑEIRO, Verónica; DAMILANO, Bianca, in the National University of the South (from the Spanish: Universidad Nacional del Sur).

Photos of the presentation at 02/03/2023 are shown in Figures 4, 5, 6 and 7. Other documents related to this event and the interviews published, are presented below:



Coronel Suárez, Febrero 2023

Por la presente, tenemos el agrado de comunicarles la inauguración de nuestra planta de Biogás en el Establecimiento "La Esquina", ubicado sobre Ruta Provincial 85, km 166.

Es un momento muy importante para nuestro equipo de trabajo, y motivo suficiente para contar con su inestimable presencia en el evento.

El acto inaugural se llevara a cabo el jueves 2 de marzo de 2023, a las 11 hs.
Luego compartiremos un almuerzo campestre.

Por favor, confirmar asistencia a biodes@agrodesouza.com.ar

Un saludo cordial.

Mario L. y Cecilio de Souza - Javier y Cristian Haack



Figure 4. Invitation to the public presentation



Figure 5. BiodeS administration presenting the event



Figure 6. BiodeS plant



Figure 7. BiodeS project's presentation



Figure 8. BiodeS project installations

Date	Entity	Interviewer	Media	Link
1/9/2019	El Campo, La Industria Verde	Héctor Huergo	YouTube	https://www.youtube.com/watch?v=JaB0VhwLa8I&t=221s \t " _blank
20/3/2021	Clarín	Esteban Fuentes	Online diary	El feedlot marca el camino de la sustentabilidad: producir carne y energía
18/4/2021	Por La Vida TV	Jorge Romano	YouTube	AGRO DE SOUZA
11/12/2021	La Nueva	Lucía Rodríguez	Online diary	Coronel Suárez: construye una planta de biogás para transformar en energía los desechos de un feed-lot
11/4/2022	Municipio Cnel. Suarez		Instagram	https://www.instagram.com/tv/CcOkxMoFme-/
19/6/2022	La Calicata	Patricia Chimeno	AM 1240 - Spotify	Adaptación de un sistema ganadero en el partido de Coronel Suarez: feed lot estabulado y planta de Biogás
15/8/2022	La Calicata	Patricia Chimeno	AM 1240 - Spotify	Las oportunidades del Biogas: Visita a Italia para conocer las experiencias del biogas
9/11/2022	A corral	Delsector Grupo Guarino	YouTube	A corral 09/11/22 - YouTube
16/11/2022	A corral	Delsector Grupo Guarino	YouTube	A corral 16/11/22 - YouTube
4/2/2023	Producción Coronel Suárez		YouTube	Produccion 703 04 02 2023 - YouTube
25/2/2023	Clarín	Esteban Fuentes	Online diary	Construyó un feedlot con piso de hormigón y techo y con la bosta de los animales genera electricidad para 2.000 personas (clarin.com)
8/3/2023	Portal Agropecuario		Online diary / Radio y RRSS	Portal Agropecuario " Con la nueva planta de biogás cerramos el circuito de una economía circular donde ganamos todos "
4/3/2023	Economía Sustentable		Online diary	Así es la nueva planta que transforma en energía desechos de un feed-lot en Coronel Suárez (economiasustentable.com)
13/3/2023	Por La Vida TV	Jorge Romano	YouTube	https://www.youtube.com/watch?v=jow9zBPt-54

Table 3. Interviews published

2.3 Environmental Impact

The environmental matters in Argentina are jurisdiction of the Provincial Governments. The Government of Buenos Aires sanctioned the Provincial Law of Protection, Conservation, Improvement and Restoration of Natural Resources and the Environment in general (Law No. 11,723) on 22/12/1995³. The Resolution 510/2018, in Article 10° of Law No. 11,723 establishes that all projects consisting of activities that produce or are likely to produce any negative effect to the environment of the Province of Buenos Aires and/or its natural resources, must obtain an EIA.

Additionally, as mentioned in 1.14 above, in the context of the Argentinean electricity market, all new agents (generators, self-generators, co-generators or transmission agent) that aim to participate in the MEM, must present the request to be a MEM agent in conjunction with the EIA to the Argentinean Ministry of Energy⁴.

According to these requirements, the project developer elaborated the EIA mentioned in section 2.1 above and submitted it to the Buenos Aires Province Government and to the Ministry of Energy of the Nation, complying with the requirements of National Electricity Regulatory Entity Resolution No. 13/97⁵.

The Project meets all national law and regulations since the Buenos Aires Province Government approved the Resolutions N° “IF-2022-31332861-GDEBA-DPEIAMAMGP” the 19th of September of 2022 and N° “RESO-2022-565-GDEBA-SSCYFAMAMGP” the 2nd of December of 2022.

2.4 Public Comments

This section is not required to be filled before the project undergoes a public comment period.

2.5 AFOLU-Specific Safeguards

The project is a non-AFOLU project. For non-AFOLU projects, this section is not required.

³ Law No. 11,227 - Provincial Law of Protection, Conservation, Improvement and Restoration of Natural Resources and the Environment in general <https://normas.gba.gob.ar/ar-b/ley/1995/11723/5117#:~:text=Normas%20relacionadas%20con%20Ley%2011723&text=ESTABL ECE%20LA%20CAPACITACI%C3%93N%20OBLIGATORIA%20EN,A%20ADHERIR%20A%20LA %20PRESENTE>

⁴ Environmental Requirements for Entry into the Wholesale Electricity Market <http://www.energia.gob.ar/contenidos/verpagina.php?idpagina=3433>

⁵ Resolution ENRE 0013/1997 [https://www.enre.gov.ar/web/bibliotd.nsf/\(\\$IDWeb\)/132C161BD50E954E862564210069D752](https://www.enre.gov.ar/web/bibliotd.nsf/($IDWeb)/132C161BD50E954E862564210069D752)

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The methodologies applied are both CDM small scale methodologies referred as

- AMS-III.D - Methane recovery through controlled anaerobic digestion (version 21.0)⁶.

This methodology also refers to the latest version of the following tools:

- Tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0)⁷.
- Tool 14: Project and leakage emissions from anaerobic digesters (version 02.0)⁸
- AMS-I.D. - Grid connected renewable electricity generation (version 18.0)⁹.

This methodology also refers to the latest version of the following tools:

- Tool 07: Tool to calculate the emission factor for an electricity system (version 07.0)¹⁰.

3.2 Applicability of Methodology

The project satisfies all the applicability criteria of the methodology AMS-III.D (Version 21.0), of which the detailed description is listed in the table below:

⁶ AMS-III.D. <https://cdm.unfccc.int/methodologies/DB/H9DVSB24O7GEZQYLYNWUX23YS6G4RC>

⁷ Tool 5 - version 03.0 <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v3.0.pdf>

⁸ Tool 14 - version 02.0 <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-14-v2.pdf>

⁹ AMS-I.D. - version 18.0 <https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTXFQQOFQQH4SBK>

¹⁰ Tool 07 - version 07.0 <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

No.	Applicability criteria	Project compliance with the applicability condition
2	This methodology covers project activities involving the replacement or modification of anaerobic animal manure management systems in livestock farms to achieve methane recovery and destruction by flaring/combustion or gainful use of the recovered methane. It also covers treatment of manure collected from several farms in a centralized plant.	Applicable. The project replaces an anaerobic lagoon with a new anaerobic manure management system that consists of two biodigesters. These are located in the middle of 3 different barns that provide manure to the system. The new equipment recovers and destroys methane through combustion and renewable energy production
3	This methodology is only applicable under the following conditions:	
	(a) The livestock population in the farm is managed under confined conditions;	Applicable. All the livestock population in the farm, within the project boundaries, are under confined conditions, in 24 paddocks
	(b) Manure or the streams obtained after treatment are not discharged into natural water resources (e.g., river or estuaries), otherwise "AMS-III.H Methane recovery in wastewater treatment" shall be applied;	Applicable. The digestate that results from the process is separated into solids (biosol) and liquid (biol). Part of the liquid returns to the process and the rest is used for soil application as well as the solid part.
	(c) The annual average temperature of baseline site where anaerobic manure treatment facility is located is higher than 5°C;	Applicable. The project site is in Buenos Aires Province where the annual average temperature is 17°C. The minimum annual average of the coldest month in the year 2022 was 9°C.
	(d) In the baseline scenario the retention time of manure waste in the anaerobic treatment system is greater than one month, and if anaerobic lagoons are used in the baseline, their depths are at least 1 m;	Applicable. it is assumed that the anaerobic lagoon defined as the baseline scenario is more than 1 m depth and wastewater is retained for more than 1 month.
	(e) No methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario.	Applicable. In the baseline scenario, there is no methane recovered for any gainful use.

No.	Applicability criteria	Project compliance with the applicability condition
4	The project activity shall satisfy the following conditions:	
	(a) The residual waste from the animal manure management system shall be handled aerobically, otherwise the related emissions shall be taken into account as per relevant procedures of "AMS-III.AO Methane recovery through controlled anaerobic digestion". In the case of soil application, proper conditions, and procedures (not resulting in methane emissions) must be ensured;	Applicable. The solid fraction of the waste is composted aerobically and then spread as a biofertilizer on the soil. The liquid fraction is conducted to a deep open lagoon to be used as an effluent diluent in the loading channels and for irrigation. These emissions will be considered.
	(b) Technical measures shall be used (including a flare for exigencies) to ensure that all biogas produced by the digester is used or flared;	Applicable. The biogas produced will be used for energy production by the motor generator. If it cannot be used in the cogeneration unit due to maintenance work, it will be burned in the flaring torch.
	(c) The storage time of the manure after removal from the animal barns, including transportation, should not exceed 45 days before being fed into the anaerobic digester. If the project proponent can demonstrate that the dry matter content of the manure when removed from the animal barns is larger than 20%, this time constraint will not apply.	Applicable. The digester is fed daily, so the retention time of the manure in the loading chamber is less than 45 days. In addition, the percentage of dry matter in manure collected is determined as 27,27% through a laboratory analysis. This evaluation will be available for the audit team.
5	Projects that recover methane from landfills shall use "AMS-III.G Landfill methane recovery" and projects for wastewater treatment shall use AMS-III.H. Projects for composting of animal manure shall use "AMS-III.F Avoidance of methane emissions through composting". Project activities involving co-digestion of animal manure and other organic matters shall use the methodology "AMS-III.AO Methane recovery through controlled anaerobic digestion".	Not applicable. This project activity does not meet any of the conditions described in this section.

No.	Applicability criteria	Project compliance with the applicability condition
6	Utilization of the recovered biogas in one of the options detailed in AMS-III.H is also eligible under this methodology. The respective procedures in AMS-III.H shall be followed in this regard. If the recovered biogas is used to power auxiliary equipment of the project activity, it should be taken into account, accordingly, using zero as its emission factor; however, energy used for such purposes is not eligible as an SSC CDM Type I project component.	Not applicable. This project does not involve the recovery of methane from wastewater.
7	New facilities (Greenfield projects) and project activities involving capacity additions compared to the baseline scenario are only eligible if they comply with the related and relevant requirements in the "General guidelines for SSC CDM methodologies ¹¹ ".	Applicable. The project consists of new facilities and complies with all the related and relevant requirements in the "General guidelines for SSC CDM methodologies".
8	The requirements concerning demonstration of the remaining lifetime of the replaced equipment shall be met as described in the "General guidelines for SSC CDM methodologies".	Not applicable. The project consists of a Greenfield project with no replacement of any equipment. All facilities are new.
9	Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO ₂ equivalent annually from all Type III components of the project activity.	Applicable. The average ERs for this project are 14,399 t CO ₂ e annually, being less than 60,000 t CO ₂ e per year.

The project also satisfies all the applicability criteria of the Tools 05 and 14 of which the detailed description is listed in the table below:

¹¹ General guidelines for SSC CDM methodologies -
https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210211212225226/MethSSC_Guid25ver23.1.pdf

Tool	Applicability criteria	Project compliance with the applicability condition
Tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (Version 03.0)	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer. (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	Applicable for scenario A. Electricity in the project is consumed from the national grid and from a solar panel source installed in the roofs of feedlot paddocks.
	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	Applicable for scenario A. Electricity generated in the biogas plant is 100% supplied to the national grid.

Tool	Applicability criteria	Project compliance with the applicability condition
	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO ₂ emissions.	Applicable. The energy produced in the biogas plant is not used for the project activity.
Tool 14: Project and leakage emissions from anaerobic digesters (version 2.0)	The following sources of project emissions are accounted for in this tool: (a) CO ₂ emissions from consumption of electricity associated with the operation of the anaerobic digester; (b) CO ₂ emissions from consumption of fossil fuels associated with the operation of the anaerobic digester; (c) CH ₄ emissions from the digester (emissions during maintenance of the digester, physical leaks through the roof and side walls, and release through safety valves due to excess pressure in the digester); and (d) CH ₄ emissions from flaring of biogas.	Applicable for scenarios A and C. In this case, the emissions considered will be the ones due to the consumption of electricity for operations and due to physical leaks and releases through safety valves. When biodigesters need maintenance, the flaring torch will be used.
	The following sources of leakage emissions are accounted for in this tool: (a) CH ₄ and N ₂ O emission from composting of digestate. (b) CH ₄ emissions from the anaerobic decay of digestate disposed in a SWDS or subjected to anaerobic storage, such as in a stabilisation pond.	Applicable for scenario B. The liquid digestate is left to decay in an anaerobic lagoon part of the year. The solid part is composted aerobically, so emissions are not accounted for.
	Emission sources associated with N ₂ O emissions from physical leakages from the digester, transportation of feed material and digestate or any other on-site transportation, piped distribution of the biogas, aerobic treatment of liquid digestate and land application of the digestate are neglected because these are minor emission sources or because they are accounted in the methodologies referring to this tool.	Applicable. As per the applied methodology, N ₂ O emissions are neglected because these are minor emission sources.

See the project applicability to the AMS-I.D (Version 18.0) methodology, of which the detailed description is listed below:

Paragraph No.	Applicability criteria	Project compliance with the applicability condition
4	This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s).	Applicable for Scenario (a). The present project activity is a greenfield biogas power plant.
5	Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m²; (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m².	Not applicable. The present project activity is not a hydro power plant.
6	If the new unit has both renewable and non-renewable components (e.g., a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	Not applicable. The new unit has only renewable components.
7	Combined heat and power (co-generation) systems are not eligible under this category.	Applicable. The project activity is not a co-generation system. It produces simultaneously electrical and thermal energy, but this last one is recovered to transfer heat to water used to maintain the

Paragraph No.	Applicability criteria	Project compliance with the applicability condition
		required temperature in the mesophilic reactors.
8	In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	Not applicable. The present project activity is a greenfield power plant.
9	In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW.	Not applicable. The present project activity is a greenfield power plant.
10	In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as “AMS-I.C.: Thermal energy production with or without electricity” shall be explored.	The project consists of an agroindustry project, where recovered methane emissions (eligible under AMS-III.D as demonstrated above) are used for electricity generation, supplied to the national grid. The baseline for the electricity component is in accordance with procedure prescribed under AMS-I.D. This establishes that the electricity delivered to the grid by the project activity, at a baseline scenario, would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid (paragraph 19).

Paragraph No.	Applicability criteria	Project compliance with the applicability condition
11	In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply.	Not applicable. The biomass is not sourced from dedicated plantations.

The project also satisfies all the applicability criteria of the Tool 07 of which the detailed description is listed in the table below:

Tool	Applicability criteria	Project compliance with the applicability condition
Tool 07: Tool to calculate the emission factor for an electricity system (version 07.0)	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	Applicable. The project consists in the substitution of grid electricity by a renewable source.
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 percent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 percent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Applicable. The present project consists of a power plant connected to the grid. The emission factor is calculated based in grid power plants only.
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Argentina is not an Annex

Tool	Applicability criteria	Project compliance with the applicability condition
		I country ¹² .
	Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	Applicable. Emission of CO ₂ derive from biomass burning so the emission factor is considered 0.

3.3 Project Boundary

According to the methodology AMS-III.D, the project boundary includes the physical, geographical site(s) of:

- (a) The livestock;
- (b) Animal manure management systems (including centralised manure treatment plant where applicable);
- (c) Facilities which recover and flare/combust or use methane.

Additionally, AMS-I.D (version 18.0) methodology states in paragraph 18 that the spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the project power plant is connected to.

Therefore, the sites and processes that are part of the project activities and are considered within the project boundaries can be seen in the figure below, in accordance with both methodologies:

¹² List of Annex I countries - <https://unfccc.int/process/parties-non-party-stakeholders/parties-convention-and-observer-states>

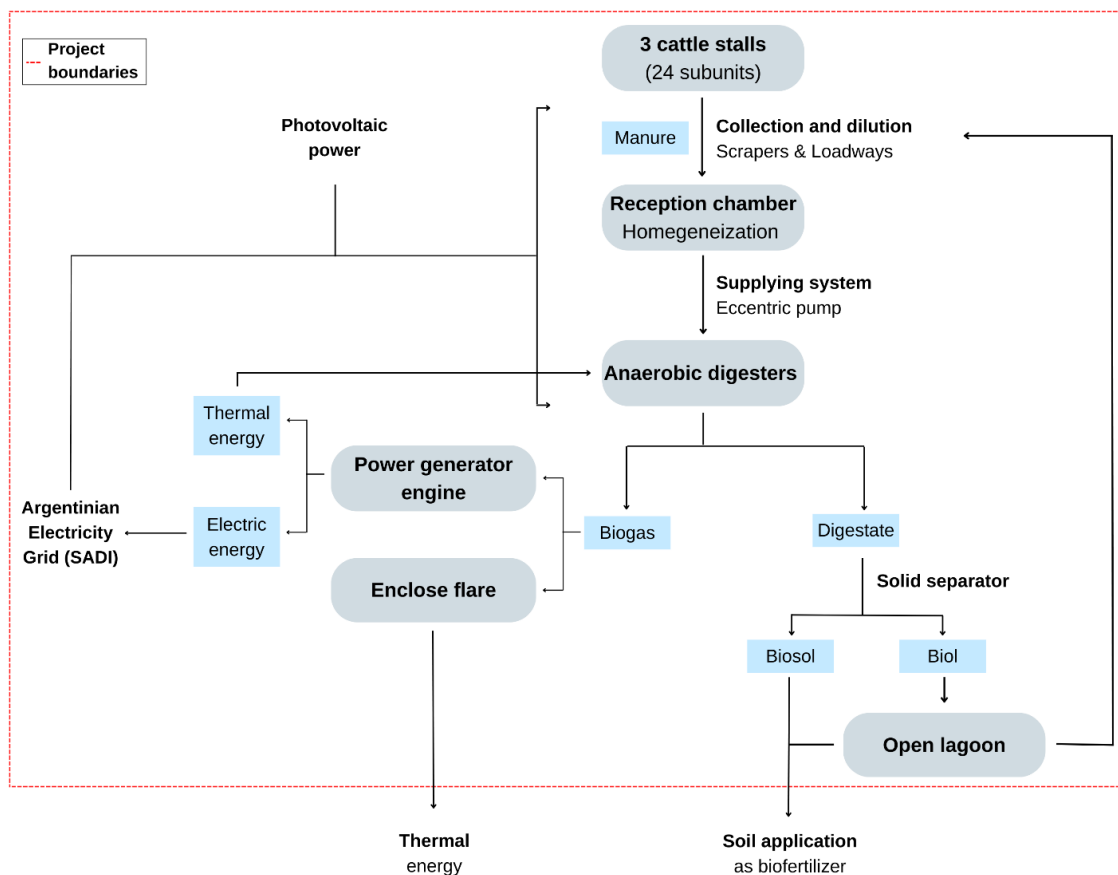


Figure 9. BiodeS process map

The emissions sources included or excluded from the project boundary are summarised in the following table.

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	No	Excluded for simplification.
	CH ₄	Yes	The major source of emissions in the baseline.
	N ₂ O	No	Excluded for simplification.
Project	CO ₂	No	Excluded for simplification.
	CH ₄	Yes	Main emission source.
	N ₂ O	No	Excluded for simplification.

Source	Gas	Included?	Justification/Explanation	
	Emissions from flaring or combustion of the gas stream	CO ₂	No	Excluded for simplification.
		CH ₄	No	Excluded for simplification. The flaring system in the project activity is installed for emergencies, so the emissions are assumed to be negligible. The project will not claim this emission reductions
		N ₂ O	No	Excluded for simplification.
	Emissions from use of fossil fuels or electricity	CO ₂	Yes	The baseline scenario already includes the use of a fossil-fuel engine for electricity generation for the irrigation of water to the farm. In the project this activity remains constant, so the emissions are not considered. The emissions due to electricity used by the feedlot functioning will not be considered as project emissions because they would already exist in the absence of the project. On the other hand, electricity consumed by the biogas plant will be considered in the net energy generated and delivered to the grid. This will determine ER’s claimed.
		CH ₄	No	Excluded for simplification.
		N ₂ O	No	Excluded for simplification.
	Emissions from incremental transportation distances	CO ₂	No	Excluded for simplification.
		CH ₄	No	There is no incremental transportation in the project.
		N ₂ O	No	Excluded for simplification.
	Emissions	CO ₂	No	Excluded for simplification.

Source	Gas	Included?	Justification/Explanation
from storage of manure	CH ₄	No	The storage time of the manure after removal from the animal barns is within 24 hours before being fed into the anaerobic digester, hence emissions from the storage of manure are not accounted for.
	N ₂ O	No	Excluded for simplification.

3.4 Baseline Scenario

Paragraph 17 of AMS-III.D defines the baseline scenario is the situation where, in the absence of project activities, animal manure decays anaerobically within the project boundary and methane is emitted to the atmosphere.

Therefore, the baseline scenario of this project is that the manure of livestock is left to decay in an open lagoon (anaerobic manure management system), and the methane is directly emitted to the atmosphere without any methane recovery nor destruction facilities.

As per AMS-I.D methodology, paragraph 19, the project's baseline scenario for greenfield power plants is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

3.5 Additionality

As per paragraph 15-16 of AMS-III.D, project activities may demonstrate the additionality by showing that there is no regulation in the host country, applicable to the project site that requires the collection and destruction of methane from livestock manure. If so, it is not required to apply the "Guidelines on the demonstration of additionality of small-scale project activities". This additionality condition also applies to Greenfield project activities. Furthermore, for project activities applying this methodology in combination with a Type I methodology, that has an energy component whose installed capacity is less than 5 MW, this procedure for additionality demonstration also applies to that component.

At a national level mainly, this is demonstrated in Resolution N° 70/2001 and 329-E/2017¹³ of the "National Registry of Feedlot Livestock Establishments" and the Resolution N°

¹³ Resolution N° 70/2001 and 329-E/2017 -

<https://www.argentina.gob.ar/normativa/nacional/resoluci%C3%B3n-329-2017-274944/texto>

097/01¹⁴ of the Ministry of Social Development and Environment. In these the government defines regulations for the preservation of the environment in the generation of livestock effluents. However, it does not specify the construction of a biodigester, where the most common practice is the development of an anaerobic open lagoon.

All the above Argentinian laws and regulations do not demand the recovery and destruction of methane from livestock manure. In line with AMS-III.D, the project is considered automatically additional. Furthermore, as it was stated before in title 3.2, Argentina is not an Annex 1 country.

3.6 Methodology Deviations

No methodology deviations.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Baseline emissions are calculated separately in the two components this project has, as per each methodology defines the calculations: Methane recovery and renewable energy generation.

According to methodology AMS-III.D, Baseline Emissions ($BE_{I,y}$) for the methane recovery component can be calculated by using two options (paragraph 17). In the case of this project option (a) is used, which is described below:

- (a) Using the amount of the waste or raw material that would decay anaerobically in the absence of the project activity, with the most recent IPCC Tier 2 approach (please refer to the chapter 'Emissions from Livestock and Manure Management' under the volume '*Agriculture, Forestry and other Land use*' of the *2006 IPCC Guidelines for National Greenhouse Gas Inventories*). For this calculation, information about the characteristics of the manure and of the management systems in the baseline is required. Manure characteristics include the amount of volatile solids (VS) produced by the livestock and the maximum amount of methane that can be potentially produced from that manure (B_0);

¹⁴ Resolution N° 097/01 - <https://www.argentina.gob.ar/normativa/nacional/resoluci%C3%B3n-97-2001-70192>

It is necessary to clarify that, as the methodology was presented at CDM before 2019 IPCC refinement, it refers to the 2006 guidelines for reference. However, for this project 2019 refinement will be used because they are updated and improved from 2006.

Baseline emissions (BE_y) are determined as follows:

$$BE_y = GWP_{CH_4} \times D_{CH_4} \times UF_b \times MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{BL,j}$$

Where:

BE_y = Baseline emissions in year y (t CO₂e)

GWP_{CH_4} = Global Warming Potential (GWP) of CH₄ applicable to the crediting period (t CO₂e/t CH₄)

D_{CH_4} = CH₄ density (0.00067 t/m³ at room temperature (20 °C) and 1 atm pressure)

LT = Index for all types of livestock

j = Index for animal manure management system

MCF_j = Annual methane conversion factor (MCF) for the baseline animal manure management system j

$B_{0,LT}$ = Maximum methane producing potential of the volatile solid generated for animal type LT (m³ CH₄/kg-dm)

$N_{LT,y}$ = Annual average number of animals of type LT in year y (numbers)

$VS_{LT,y}$ = Volatile solids production/excretion per animal of livestock LT in year y (on a dry matter weight basis, kg-dm/animal/year)

$MS\%_{BL,j}$ = Fraction of manure handled in baseline animal manure management system j

UF_b = Model correction factor to account for model uncertainties (0.94)¹⁵

In paragraph 18 of the methodology, it is described each component of equation 1. In section (a) of this paragraph, it explains that the maximum methane-producing capacity of the manure (B_0) varies by species and diet. The methodology gives two options, use data from country-specific published sources, or default values from tables 10 A-4 to 10 A-9 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 can

¹⁵ FCCC/SBSTA/2003/10/Add.2, page 25 -

<https://unfccc.int/resource/docs/2003/sbsta/10a02.pdf>

be used, provided that the project participants assess the suitability of those data to the specific situation of the treatment site. This was the method used to determine the B_o value, however, as it is mentioned before, the IPCC 2019 refinement was used, where table 10.16 defines the value for high productivity systems of non-dairy cattle.

In section (b) VS is described as the organic material in livestock manure and consists of both, biodegradable and non-biodegradable fractions. For the calculations the total VS excreted by each animal species is required, so BiodeS takes samples from manure and sends it to BGA laboratory in Argentina, where this develops the estimation through a loss-on-ignition method. This laboratory study will be available for auditors' consultation.

As the methodology states, these results shall be compared with IPCC default values, where BiodeS study is significantly higher. As it is common in the region of the project (Coronel Suarez), producers normally give cattle a high fibre content diet in feedlots. Bibliography explains that this nutrient allows a better ruminal function and solidifies manure, increasing the percentage of solids in it¹⁶. All diets supplied at BiodeS include a multiplier and modulator additive of the ruminal microbiota called Biopremix¹⁷. Studies show that this multienzyme complex increases digestibility of high fibre diets, improving the results. This information will be available for the VVB (Validation & Verification Body).

Methane Conversion Factors (MCF) values are determined for a specific manure management system and represent the degree to which B_o is achieved. Where available country specific MCF values that reflect the specific management systems used in particular countries or regions shall be used. Alternatively, the IPCC default values provided in table 10.17 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 can be used. In IPCC 2019, the MCF is determined in table 10.17 for an open lagoon in a warm temperate moist climate as is the location of the project in Argentina.

To continue, the annual average number of animals ($N_{LT,y}$) is determined as follows:

$$N_{LT,y} = N_{da,y} \times \left(\frac{N_{p,y}}{365} \right)$$

¹⁶ Manure study and it's relation to nutrition - https://www.produccion-animal.com.ar/informacion_tecnica/manejo_del_alimento/61-heces_del_bovino_y_relacion_con_la_alimentacion.pdf

¹⁷ Biopremix - https://patentscope.wipo.int/search/es/detail.jsf?docId=WO2019102279&_cid=P20-LE2Y52-29858-1

Where:

$N_{da,y}$ = Number of days animal is alive in the farm in the year y (numbers)

$N_{p,y}$ = Number of animals produced annually of type LT for the year y (numbers)

Regarding the renewable energy generation, baseline emissions according to AMS-I.D version 18.0 include only CO₂ emissions due to the electricity generation in power plants that are displaced due to the project's activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

Where:

BE_y = Baseline emissions in year y (t CO₂)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

$EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO₂/MWh)

The calculation of $EG_{PJ,y}$ is different for greenfield plants, capacity additions, retrofits, and replacements. The project activity includes installing and operating a new power generation unit. Therefore, according to the methodology, the $EG_{PJ,y}$ for a greenfield power plant is calculated as follows:

$$EG_{PJ,y} = EG_{PJ,facility,y}$$

Where:

$EG_{PJ,facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh)

Paragraph 23 of the AMS-I.D version 18.0 methodology, the emission factor shall be calculated in a transparent and conservative manner, and it suggests Tool 07 for its calculation. Here option (a) is chosen, where the combined margin (CM) is calculated by the combination of the operating margin (OM) and build margin (BM) according to the procedures prescribed.

According to Tool 07 version 07.0, project participants shall apply the following six steps:

- (a) Step 1: Identify the relevant electricity systems;
- (b) Step 2: Choose whether to include off-grid power plants in the project electricity system (optional);
- (c) Step 3: Select a method to determine the operating margin (OM);
- (d) Step 4: Calculate the operating margin emission factor according to the selected method;
- (e) Step 5: Calculate the build margin (BM) emission factor;
- (f) Step 6: Calculate the combined margin (CM) emission factor.

The six steps are detailed including the choices made applying the tool.

(a) Step 1: Identify the relevant electricity systems

The Argentinian Interconnection System (SADI – by its initials in Spanish) is the relevant electricity system.

Any imports will come from connected electricity systems located outside of Argentina. Therefore, following the guidance given by the Tool, an emission factor of 0 t CO₂ per MWh is chosen for electricity imports.

(b) Step 2: Choose whether to include off-grid power plants in the project electricity system

For this step, option 1 was chosen, where only grid power plants are included in the calculation, as off grid power generation is not significant.

(c) Step 3: Select a method to determine the operating margin (OM)

The Tool gives four options to calculate OM. For this project, option (a) was chosen, or Simple OM, since low cost/must-run (hydro, solar, biomass and wind) resources (LCMR) accounted for are less than 50% of total grid generation over the five most recent years, according to the available information (2017-2021) as below demonstrated.

To determine the share of LCMR generation, approach 1 of paragraph 40 of the Tool indicates the following calculation:

$$Share_{LCMR} = average \left[\frac{EG_{LCMRy-4}}{total_{y-4}}, \dots, \frac{EG_{LCMRy}}{total_y} \right]$$

Where:

$Share_{LCMR}$ = Share of the low cost/must run resources (per cent)

EG_{LCMRy} = Electricity generation supplied to the project electricity system by the low cost/must run resources in year y (MWh)

$total_y$ = Total electricity generation supplied to the project electricity system in year y (MWh)

Y = The most recent year for which data is available

Cammesa¹⁸ is the organization in charge of operating the SADI, plan energy capacity needs, coordinate dispatch operations and regulate economic transactions in the wholesale electricity market.

Table 4. Historic electricity generation in Argentina by Cammesa

Electricity generation	2017	2018	2019	2020	2021
	GWh/year				
Thermal	88,531.11	87,727.04	80,137.00	82,335.51	90,074.35
Hydraulic	39,584.00	39,951.65	35,370.16	29,093.03	24,116.32
Nuclear	5,716.23	6,452.92	7,927.08	10,010.88	10,169.69
Renewable (not hydraulic)	2,634.75	3,350.23	7,812.00	12,737.16	17,436.62
Total generation	136,466.09	137,481.83	131,246.24	134,176.58	141,796.98
LCMR generation	47,934.98	49,754.80	51,109.24	51,841.07	51,722.63
% LCMR generation	35.1%	36.2%	38.9%	38.6%	36.5%
5 years average low-cost/must-run % of SADI total electricity generation					37.1%

In conclusion, the operating margin emission factor ($EF_{grid,OM,y}$) for the SADI is calculated ex ante for the Simple OM method, according to paragraph 42 (a), where the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required. For grid power plants, a 3-year generation-weighted average is used, based on the most recent data available.

(d) Step 4: Calculate the operating margin emission factor according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (t CO₂/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units.

Option B of paragraph 47 of the Tool was chosen, where the simple OM calculation is based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system. This option is selected given that:

(i) The necessary data for Option A is not available.

¹⁸ Historic data available at: <https://cammesaweb.cammesa.com/informe-anual/>

(ii) Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and

(iii) Off-grid power plants are not included in the calculation

As per paragraph 54 of the methodology, the simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low-cost/must-run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{grid,OMsimple,y} = \frac{\sum_i FC_{i,y} \times NCV_{i,y} \times EF_{CO2,i,y}}{EG_y}$$

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO2 emission factor in year y (t CO ₂ /MWh)
$FC_{i,y}$	=	Amount of fuel type i consumed in the project electricity system in year y (mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO2,i,y}$	=	CO2 emission factor of fuel type i in year y (t CO ₂ /GJ)
EG_y	=	Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must-run power plants/units, in year y (MWh)
i	=	All fuel types combusted in power sources in the project electricity system in year y
y	=	The relevant year as per the data vintage chosen in Step 3

(e) Step 5: Calculate the build margin (BM) emission factor

In terms of vintage of data, Option 1 was chosen. For the first crediting period, the calculation of the build margin emission factor was determined ex ante, based on the most recent information available on units already built for sample group m at the time of CDM-PDD submission to the DOE for validation. The most recent information available of SADI is 2021 data.

As per Tool 07.0 Version 07.0 capacity additions from retrofits of power plants are not included in the calculation of the build margin emission factor, and the sample group of power units m used to calculate the build margin was determined as it states in paragraph 75. As a result, in 2021 the SET $\geq$ 20% had a larger production than SET5 units, so SET $\geq$ 20% has been considered the sample set (SET sample).

For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.

The build margin emissions factor is the generation-weighted average emission factor (t CO₂/MWh) of all power units m during the most recent year y for which electricity generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}}$$

Where:

$EF_{grid,BM,y}$	=	Build margin CO ₂ emission factor in year y (t CO ₂ /MWh)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (t CO ₂ /MWh)
m	=	Power units included in the build margin
y	=	Most recent historical year for which electricity generation data is available

$EF_{EL,m,y}$ of each power plant was calculated according to Option A1 selected in Step 4.

(f) Step 6: Calculate the combined margin (CM) emission factor

The calculation of the CM emission factor ($EF_{grid,CM,y}$) is based on the weighted average method (Option a, paragraph 81), which is the preferred method as indicated by the Tool. The CM emission factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times W_{OM} + EF_{grid,BM,y} \times W_{BM}$$

Where:

$EF_{grid,BM,y}$	=	Build margin CO ₂ emission factor in year y (t CO ₂ /MWh)
$EF_{grid,OM,y}$	=	Operating margin CO ₂ emission factor in year y (t CO ₂ /MWh)
W_{OM}	=	Weighting of operating margin emissions factor (per cent)
W_{BM}	=	Weighting of build margin emissions factor (per cent)

4.2 Project Emissions

According to the methodology AMS-III.D, project emissions consist of:

- Physical leakage of biogas in the manure management systems which includes production, collection and transport of biogas to the point of flaring/combustion or gainful use ($PE_{PL,y}$);

- (b) Emissions from flaring or combustion of the gas stream ($PE_{flare,y}$);
- (c) CO₂ emissions from use of fossil fuels or electricity for the operation of all the installed facilities ($PE_{power,y}$);
- (d) CO₂ emissions from incremental transportation distances ($PE_{transp,y}$);
- (e) Emissions from the storage of manure before being fed into the anaerobic digester ($PE_{storage,y}$).

$$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y}$$

Where:

PE_y = Project emissions in year y (t CO₂e)

$PE_{PL,y}$ = Emissions due to physical leakage of biogas in year y (t CO₂e)

$PE_{flare,y}$ = Emissions from flaring or combustion of the biogas stream in the year y (t CO₂e)

$PE_{power,y}$ = Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO₂e)

$PE_{transp,y}$ = Emissions from incremental transportation in the year y (t CO₂e), as per relevant paragraph in AMS-III.AO

$PE_{storage,y}$ = Emissions from the storage of manure (t CO₂e)

In the case of this project, the emissions that will be considered are the $PE_{PL,y}$ and $PE_{power,y}$. The project does not present emissions for the other parameters. Each one of these is explained below.

(a) Emissions due to physical leakage of biogas in year y

Project emissions due to physical leakage of biogas from the animal manure management systems used to produce, collect, and transport the biogas to the point of flaring or gainful use are estimated through Tool 14 "Project and leakage emissions from anaerobic digesters". In such a case, $PE_{PL,y}$ is equivalent to $PE_{CH_4,y}$ in the tool.

The Tool defines that project emissions of methane from the anaerobic digester include emissions during maintenance of the digester, physical leaks through the roof and side walls, and release through safety valves due to excess pressure in the digester. These emissions are calculated using a default emission factor ($EF_{CH_4, default}$), as follows:

$$PE_{CH_4,y} = Q_{CH_4,y} \times EF_{CH_4, default} \times GWP_{CH_4}$$

Where:

$PE_{CH_4,y}$ = Project emissions of methane from the anaerobic digester in year y (t CO₂ e)

$Q_{CH_4,y}$ = Quantity of methane produced in the anaerobic digester in year y (t CH₄)

$EF_{CH_4,default}$ = Default emission factor for the fraction of CH₄ produced that leak from the anaerobic digester (fraction)

GWP_{CH_4} = Global warming potential of CH₄ (t CO₂ / t CH₄)

Section 6.3 determines the default values for parameters established in equations. Table 3 describes parameter $EF_{CH_4,fault}$ in reference to the type of digester used in the project activity. This project corresponds to steel digesters and a gas holding system (egg shaped digesters) and a monolithic construction.

(b) Emissions from flaring or combustion of the biogas

As per par. 22 of AMS-III.D, if the recovered biogas is combusted for electrical/thermal energy production or for other gainful use, the methane destruction efficiency can be considered as 100%. The project activity uses the recovered biogas for energy production, and it has an enclosed flare installed for emergency situations. As a result, the project activity is not expected to utilise the emergency flare to combust excess biogas containing methane.

Therefore, ex ante no methane emissions due to incomplete flaring are expected:

$$PE_{flare,y} = 0$$

However, this use of the recovered biogas is included in the project boundary and its output will be monitored to ensure that the recovered biogas is destroyed.

(c) Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y

According to AMS-III.D, project emissions from electricity and fossil fuel consumption are determined by following the methodological Tool 14 “Project and leakage emissions from anaerobic digesters”, where $PE_{power,y}$ is the sum of $PE_{EC,y}$ (Project emissions from electricity consumption associated with the anaerobic digester in year y) and $PE_{FC,y}$ (Project emissions from fossil fuel consumption associated with the anaerobic digester in year y).

In the case of this project, $PE_{FC,y} = 0$, because no new fossil fuel use is added to the project in reference to the scenario prior to project implementation. Furthermore, $PE_{EC,y} =$

0 because the energy generated by the project is measured by a bidirectional electricity meter regulated by CAMMESA¹⁹. This national organism publishes a monthly report with the net energy generated, as the total energy delivered to the national grid minus the amount of energy consumed by the biogas plant.

In conclusion,

$$PE_{power,y} = 0$$

(d) Emissions from incremental transportation in the year y

The digester is installed within the Agro de Souza Farm, so there's no incremental transportation. Therefore,

$$PE_{transp,y} = 0$$

(e) Emissions from the storage of manure

Project emissions on account of storage of manure before being fed into the anaerobic digester shall be accounted for if both condition (a) and condition (b) below are satisfied:

- (a) The storage time of the manure after removal from the animal barns, including transportation, exceeds 24 hours before being fed into the anaerobic digester.
- (b) The dry matter content of the manure when removed from the animal barns is less than 20%.

The storage time of the manure after removal from the animal barns is within 24 hours before being fed into the anaerobic digester. Furthermore, the dry matter content of manure is around 27%, higher than the established limit. Therefore, emissions from the storage of manure are not accounted for. Therefore,

$$PE_{storage,y} = 0$$

4.3 Leakage

As per the Tool 14 "Project and leakage emissions from anaerobic digesters" (version 02.0), the leakage emissions associated with the anaerobic digester ($LE_{AD,y}$) depends on how the digestate is managed. These include emissions associated with storage of digestate ($LE_{storage,y}$) and its composting ($LE_{comp,y}$) (Equation 5 of the methodology).

Step 1: Determination of leakage emissions associated with storage of digestate ($LE_{storage,y}$)

¹⁹ CAMMESA - <https://cammesaweb.cammesa.com/>

Storage of digestate under anaerobic conditions can cause CH₄ emissions due to further anaerobic digestion of the residual biodegradable organic matter. The procedure for determining $LE_{storage,y}$ is distinguished for liquid digestate (biol) and solid digestate (biosol).

Emissions from liquid storage must be considered where digestate is stored in an un-aerated lagoon with a depth greater than one meter. The PP is going to irrigate the liquid digestate, however the stages of the crops must be followed so it would be necessary to stop spreading it on the plants 60 days before making the silage. In this period the liquid stored in the open lagoon is feasible to release GHGs. These emissions will be considered and calculated as per Tool 14 states in paragraph 6.2.1.1. and Option 1 was chosen for the calculations (paragraph 6.2.1.2). Therefore,

$$LE_{storage,y} = Q_{stored,y} \times P_{COD,y} \times B_0 \times MCF_p \times GWP_{CH_4}$$

Where:

$LE_{storage,y}$ = Leakage emissions associated with storage of digestate in year y (t CO₂e)

$Q_{stored,y}$ = Amount of liquid digestate stored anaerobically in year y (m³)

$P_{COD,y}$ = Average chemical oxygen demand (COD) of the liquid digestate in year y (t COD / m³)

B_0 = Maximum methane producing capacity of the COD applied (t CH₄ / t COD)

MCF = Methane conversion factor (fraction)

GWP_{CH_4} = Global warming potential of CH₄ (t CO₂ / t CH₄)

Solid emissions are considered when it is stored in a solid waste disposal site (SWDS). In the case of stockpiles, the methodology considers these a SWDS if: (a) their volume to surface area ratio is 1.5 (m³ / m²) or larger; and if (b) a visual inspection by the DOE confirms that the material is exposed to anaerobic conditions (i.e. it has a low porosity and is moist). This is not the case of BiodeS, where stockpiles are less than 1.5 m³/m², because the solid outflow is constantly applied to the soil. Thus, these emissions will not be considered.

Step 2: Determination of leakage emissions associate with composting digestate ($LE_{comp,y}$)

According to the applicability conditions of the Tool 13 "Project and leakage emissions from composting" paragraph 33, leakage emissions from composting ($LE_{comp,y}$) shall be accounted for if compost is subjected to anaerobic storage or disposed of in a SWDS. In this project the compost is managed aerobically and used as biofertilizer. Therefore, emissions corresponding to compost should not be taken into account.

$$LE_{comp,y} = 0$$

In this sense,

$$LE_{AD,y} = LE_{storage,y}$$

4.4 Net GHG Emission Reductions and Removals

As it was stated before, the present project activity reduces GHG emissions by two components: Methane recovery through anaerobic digesters and renewable energy generation from biogas.

Regarding methane recovery, the emission reduction ($ER_{1,y}$) by the project activity in year y is the difference between the baseline emissions (BE_y) and the sum of the project emissions ($PE_{1,y}$) and leakage ($LE_{1,y}$), as follows:

$$ER_{1,y} = BE_{1,y} - PE_{1,y} - LE_{1,y}$$

Baseline emissions ($BE_{1,y}$) regarding methane recovery:

$$BE_{1,y} = GWP_{CH_4} \times D_{CH_4} \times UF_b \times MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{BL,j}$$

Symbol	Unit	Value	Description	Reference
GWP_{CH_4}	t CO ₂ e / t CH ₄	28	Global Warming Potential (GWP) of CH ₄ applicable to the crediting period	IPCC AR5, as per VCS Standard Version 4.4, paragraph 3.15.4
D_{CH_4}	t/m ³	0.00067	CH ₄ density at room temperature (20 °C) and 1 atm pressure	AMS III.D Version 21.0
UF_b	-	0.94	Model correction factor to account for model uncertainties	FCCC/SBSTA/2003/10/Add. 2, page 25.
MCF_j	-	73%	Annual methane conversion factor (MCF) for the baseline animal manure management system	Default value in table 10.17 of the IPCC 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10. (Climate zone: Warm Temperate Moist)
$B_{0,LT}$	m ³ CH ₄ / kg-dm	0.209	Maximum methane producing potential of the volatile solid generated for animal type LT	Measured through laboratory analysis.

Symbol	Unit	Value	Description	Reference
$N_{LT,y}$	Numbers	3,300	Annual average number of animals of type LT in year y	Calculated. Equation (4) of AMS III.D Version 21.0
$VS_{LT,y}$	kg-dm / animal / year	1,998	Volatile solids production/excretion per animal of livestock LT in year y	Measured through laboratory analysis.
$MS\%_{BI,j}$	-	100%	Fraction of manure handled in baseline animal manure management system j	Project developer. Manure baseline system design.
$BE_{1,y}$	tCO ₂ e / year	17,739	Baseline emissions in year y	Calculated. Equation 1 of AMS III.D Version 21.0

Where:

$$N_{LT,y} = N_{da,y} \times \left(\frac{N_{p,y}}{365} \right)$$

Symbol	Unit	Value	Description	Reference
$N_{da,y}$	Numbers	191	Number of days animal is alive in the farm in the year y	Project developer. Average value of days animal is alive (between 120-210 days).
$N_{p,y}$	Numbers	6,609	Number of animals produced annually of type LT for the year y	Project developer. Value established according to the assumption that the system will be under its maximum capacity (3000 animals) throughout the year.
$N_{LT,y}$	Numbers	3,300	Annual average number of animals of type LT in year y	Calculated. Equation 4 of AMS III.D Version 21.0

Project emissions ($PE_{1,y}$) regarding methane recovery:

$$PE_{1,y} = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y}$$

As it was explained before in chapter 4.2, the PE_y correspond to the sum of $PE_{PL,y}$, $PE_{flare,y}$, $PE_{power,y}$, $PE_{transp,y}$ and $PE_{storage,y}$, where the only emissions that happen in this project are the related to physical leakage ($PE_{PL,y}$), meanwhile all the others equal 0.

Symbol	Unit	Value	Description	Reference
$PE_{flare,y}$	t CO ₂ e	0	Emissions from flaring or combustion of the biogas stream in the year y	Project design.
$PE_{power,y}$	t CO ₂ e	0	Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y	Project design.
$PE_{transp,y}$	t CO ₂ e	0	Emissions from incremental transportation	Project design.
$PE_{storage,y}$	t CO ₂ e	0	Emissions from the storage of manure	Project design.
$PE_{PL,y}$	t CO ₂ e	623	Emissions due to physical leakage of biogas in year y	Calculated. Equation 4 of Tool 14 Version 02.0
$PE_{1,y}$	tCO₂e / year	623	Project emissions in year y	Calculated. Equation 6 of AMS III.D Version 21.0

Where:

Project emissions due to physical leakage of biogas in year y ($PE_{PL,y}$) equals $PE_{CH_4,y}$ in Tool 14, and are calculated as follows:

$$PE_{CH_4,y} = Q_{CH_4,y} \times EF_{CH_4,default} \times GWP_{CH_4}$$

Symbol	Unit	Value	Description	Reference
$Q_{CH_4,y}$	t CH ₄ / year	794.51	Quantity of methane produced in the anaerobic digester in year y	Project design.
$EF_{CH_4,default}$	t CH ₄ leaked / t CH ₄ produced	0.028	Default emission factor for the fraction of CH ₄ produced that leaks from the anaerobic digester.	IPCC (2006), Flesch et al. (2011) and Kurup (2003). Digesters with steel or lined concrete or fiberglass digesters and a gas holding system (egg shaped digesters) and monolithic construction
GWP_{CH_4}	t CO ₂ e / t CH ₄	28	Global Warming Potential (GWP) of CH ₄ applicable to the crediting period	IPCC AR5, as per VCS Standard Version 4.4, paragraph 3.15.4
$PE_{PL,y}$ or $PE_{CH_4,y}$	tCO₂e / year	623	Emissions due to physical leakage of biogas in year y	Calculated. Equation 4 of Tool 14 Version 02.0

Leakage emissions ($LE_{AD,y}$) regarding methane recovery:

As per the Tool 14 “Project and leakage emissions from anaerobic digesters” (version 02.0), the leakage emissions associated with the anaerobic digester depends on how the digestate is managed. These include emissions associated with storage of digestate and its composting.

$$LE_{AD,y} = LE_{storage,y} + LE_{comp,y}$$

As it was stated in section 4.3, there are no emissions due to the composting of digestate under aerobic (oxygen-rich) conditions. Therefore,

Symbol	Unit	Value	Description	Reference
$LE_{storage,y}$	t CO _{2e}	4,070	Leakage emissions associated with storage of digestate in year y	Calculated. Equation 6 Tool 14 Version 0.20
$LE_{comp,y}$	t CO _{2e}	0	Leakage emissions associated with composting digestate in year y	Project design
$LE_{AD,y}$	tCO_{2e} / year	4,070	Leakage emissions associated with the anaerobic digester in year y	Calculated. Equation 5 Tool 14 Version 0.20

Where:

$$LE_{storage,y} = Q_{stored,y} \times P_{COD,y} \times B_0 \times MCF_p \times GWP_{CH_4}$$

Symbol	Unit	Value	Description	Reference
$Q_{stored,y}$	m ³	36,500	Amount of liquid digestate stored anaerobically in year y	Project design
$P_{COD,y}$	t COD / m ³	0.02182	Average chemical oxygen demand (COD) of the liquid digestate in year y	Project design
B_0	t CH ₄ / t COD	0.25	Maximum methane producing potential of the COD applied	Default value in table 6.2 of the IPCC 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 5 Chapter 6
MCF_p	-	73%	Annual methane conversion factor	Default value in table 10.17 of the IPCC 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter

Symbol	Unit	Value	Description	Reference
				10. (Climate zone: Warm Temperate Moist)
GWP_{CH_4}	t CO ₂ / t CH ₄	28	Global Warming Potential of CH ₄	IPCC AR5, as per VCS Standard Version 4.4, paragraph 3.15.4
$LE_{storage,y}$	tCO ₂ e / year	4,070	Leakage emissions associated with storage of digestate in year y	Calculated. Equation 6 Tool 14 Version 0.20

Emissions reduction ($ER_{1,y}$) regarding methane recovery:

Symbol	Unit	Value	Description	Reference
$BE_{1,y}$	t CO ₂ e / year	17,739	Baseline emissions in year y	Calculated. Equation 1 of AMS III.D Version 21.0
$PE_{1,y}$	t CO ₂ e / year	623	Project emissions in year y	Calculated. Equation 6 of AMS III.D Version 21.0
$LE_{1,y}$	t CO ₂ e / year	4,070	Leakage emissions in year y	Calculated. Equation 5 of Tool 14 Version 0.20
$ER_{1,y}$	t CO ₂ e / year	13,046	Emissions reductions in year y	Calculated.

Baseline emissions ($BE_{2,y}$) regarding renewable energy generation:

In reference to the renewable energy generation, AMS-I-D Version 18.0 defines,

$$BE_{2,y} = EG_{PJ,y} \times EF_{grid,y}$$

Where:

$$EG_{PJ,y} = EG_{PJ,facility,y}$$

Symbol	Unit	Value	Description	Reference
$EG_{PJ,facility,y}$	MWh	4,641	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y	Project design, Greenfield project
$EG_{PJ,y}$	MWh	4,641	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y and fed into the grid as a result of the implementation of the CDM project activity in year y	Project design, Greenfield project

Symbol	Unit	Value	Description	Reference
$EF_{grid,y}$ or $EF_{grid,CM,y}$	tCO ₂ /MWh	0.32	Combined margin CO2 emission factor for grid connected power generation in year y	Tool 07 Version 07.0 "Tool to calculate the emission factor for an electricity system"
$BE_{2,y}$	tCO ₂ e / year	1,466	Baseline emissions in year y	Calculated. Equation 1 Methodology AMS-I-D Version 18.0

The combined margin (CM) is calculated as the weighted average of the operating margin (OM) and build margin (BM) according to the procedures prescribed in the "Tool to calculate the emission factor for an electricity system" Version 07.0. Therefore,

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times W_{OM} + EF_{grid,BM,y} \times W_{BM}$$

Where:

Symbol	Unit	Value	Description	Reference
$EF_{grid,OM,y}$	tCO ₂ /MWh	0.448	Build margin CO2 emission factor in year y	Available to the VVB for validation
W_{OM}	-	50%	Weighting of operating margin emissions factor	Paragraph 86 option (b) Tool 07.0 Version 07.0
$EF_{grid,BM,y}$	tCO ₂ /MWh	0.184	Operating margin CO2 emission factor in year y	Available to the VVB for validation
W_{BM}	-	50%	Weighting of build margin emissions factor	Paragraph 86 option (b) Tool 07.0 Version 07.0
$EF_{grid,CM,y}$	tCO ₂ /MWh	0.316	Combined margin CO2 emission factor for grid connected power generation in year y	Tool 07 Version 07.0 "Tool to calculate the emission factor for an electricity system"

Project emissions ($PE_{2,y}$) regarding renewable energy generation:

AMS-I-D Version 18.0 determines that most of the renewable energy projects have a $PE_y = 0$. The only activities that have project emissions are, geothermal power plants and hydro power plants, and the current project activity is neither one of those.

As it was stated before, there is not extra consume of fossil fuels in comparison to the situation prior the project's implementation, so there are no emissions to be considered. Furthermore, the project does not involve biomass sourced from dedicated plantations. Therefore,

Symbol	Unit	Value	Description	Reference
$PE_{2,y}$	t CO ₂ e / year	0	Project emissions in year y	Project design.

Leakage emissions ($LE_{2,y}$) regarding renewable energy generation:

According to AMS-I.D Version 18.0, no leakage calculation is required for this project.
Therefore,

Symbol	Unit	Value	Description	Reference
$LE_{2,y}$	t CO ₂ e / year	0	Leakage emissions in year y	Project design.

Emissions reduction ($ER_{2,y}$) regarding renewable energy generation:

Symbol	Unit	Value	Description	Reference
$BE_{2,y}$	t CO ₂ e / year	1466	Baseline emissions in year y	Calculated. Equation 1 Methodology AMS-I-D Version 18.0
$PE_{2,y}$	t CO ₂ e / year	0	Project emissions in year y	Project design.
$LE_{2,y}$	t CO ₂ e / year	0	Leakage emissions in year y	Project design.
$ER_{2,y}$	t CO ₂ e / year	1466	Emissions reductions in year y	Calculated.

Total emission reductions (ER_y):

Considered the two sources of emission reductions of this project, the total ER_y are:

Year	Estimated baseline emissions or removals (t CO ₂ e)	Estimated project emissions or removals (t CO ₂ e)	Estimated leakage emissions (t CO ₂ e)	Estimated net GHG emissions reductions or removals (t CO ₂ e)
From 19/09/2022 to 31/12/2022	4,400	140	914	3,346
2023	19,205	623	4,070	14,512
2024	19,205	623	4,070	14,512
2025	19,205	623	4,070	14,512
2026	19,205	623	4,070	14,512

Year	Estimated baseline emissions or removals (t CO ₂ e)	Estimated project emissions or removals (t CO ₂ e)	Estimated leakage emissions (t CO ₂ e)	Estimated net GHG emissions reductions or removals (t CO ₂ e)
2027	19,205	623	4,070	14,512
2028	19,205	623	4,070	14,512
From 01/01/2029 to 18/09/2029	13,733	446	2,911	10,376
Total	133,365	4,324	28,245	100,796

5 MONITORING

5.1 Data and Parameters Available at Validation

Parameters available at validation associated to the methane recovery activity.

Data / Parameter	GWP _{CH₄}
Data unit	t CO ₂ e/t CH ₄
Description	Global Warming Potential (GWP) of CH ₄ applicable to the crediting period
Source of data	IPCC Sixth Assessment Report (AR6)
Value applied	28
Justification of choice of data or description of measurement methods and procedures applied	As per VCS Standard Version 4.4, paragraph 3.15.4
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	DCH4
Data unit	t/m3
Description	CH4 density at room temperature (20 °C) and 1 atm pressure
Source of data	AMS III.D Version 21.0
Value applied	0.00067
Justification of choice of data or description of measurement methods and procedures applied	AMS III.D Version 21.0
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	Ufb
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	FCCC/SBSTA/2003/10/Add.2, page 25.
Value applied	0.94
Justification of choice of data or description of measurement methods and procedures applied	AMS III.D Version 21.0
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	MCFj
Data unit	%
Description	Annual methane conversion factor (MCF) for the baseline animal manure management system j (j = uncovered anaerobic lagoon)

Source of data	Default value in table 10.17 of the IPCC 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10.
Value applied	73
Justification of choice of data or description of measurement methods and procedures applied	As per AMS-III.D Version 21 Climate zone: Warm Temperate Moist
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter	Bo,LT
Data unit	m3 CH4/kg-dm
Description	Maximum methane producing potential of the volatile solid generated for animal type LT
Source of data	-
Value applied	0,209
Justification of choice of data or description of measurement methods and procedures applied	Measured through laboratory analysis.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	MS%BI,j
Data unit	%
Description	Fraction of manure handled in baseline animal manure management system j
Source of data	Project developer. Manure baseline system design.
Value applied	100
Justification of choice of data or description of measurement methods and procedures applied	As per AMS-III.D Version 21
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	EFCH4,default
Data unit	t CH4 leaked / t CH4 produced
Description	Default emission factor for the fraction of CH4 produced that leaks from the anaerobic digester.
Source of data	IPCC (2006), Flesch et al. (2011) and Kurup (2003).
Value applied	0.028
Justification of choice of data or description of measurement methods and procedures applied	As per AMS-III.D Version 21 D Digesters with steel or lined concrete or fibreglass digesters and a gas holding system (egg shaped digesters) and monolithic construction
Purpose of Data	Calculation of baseline emissions
Comments	

Parameters available at validation associated to the renewable energy generation:

Data / Parameter	$EG_{m,y}$, EG_y
Data unit	MWh/year
Description	Net electricity generated by power plant/unit m ($EG_{m,y}$) or in the project electricity system (EG_y) in year y
Source of data	CAMMESA ´s website
Value applied	Data available at CAMMESA ´s annual report ²⁰
Justification of choice of data or description of measurement methods and procedures applied	Official data
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$FC_{i,m,y}$, $FC_{i,y}$
Data unit	Mass or volume unit
Description	Amount of fuel type i consumed in the project electricity system in year y
Source of data	CAMMESA ´s website
Value applied	Data available at CAMMESA ´s annual report ²⁰
Justification of choice of data or description of measurement methods and procedures applied	Official data
Purpose of Data	Calculation of baseline emissions
Comments	-

²⁰ <https://cammesaweb.cammesa.com/informe-anual/>

Data / Parameter	$NCV_{i,y}$
Data unit	GJ/mass or volume unit
Description	Net calorific value (energy content) of fuel type i in year y
Source of data	CAMMESA “Definitive Seasonal Program” 2021 ²¹
Value applied	Natural gas: 35.15 GJ/dm ³ Diesel: 43.51 GJ/m ³ Fuel Oil: 41,00 GJ/t Coal: 22.59 GJ/t
Justification of choice of data or description of measurement methods and procedures applied	Official data
Purpose of Data	Calculation of baseline emissions
Comments	-

²¹ <https://cammesaweb.cammesa.com/programacion-estacional/>

Data / Parameter	$EF_{CO2,m,i,y}$, $EF_{CO2,i,y}$
Data unit	tCO2/GJ
Description	CO2 emission factor of fuel type i used in power unit m or in the project electricity system in year y
Source of data	IPCC default values at the lower limit of the uncertainty at a 95 percent confidence interval as provided in table 1.4 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories
Value applied	Natural gas: 0.543 Diesel: 0.0726 Fuel Oil: 0.0755 Coal: 0.0873
Justification of choice of data or description of measurement methods and procedures applied	Official data
Purpose of Data	Calculation of baseline emissions
Comments	No refinement at IPCC 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories.

Data / Parameter	$EF_{grid,OMsimple,y}$
Data unit	tCO2/MWh
Description	Operating margin CO2 emission factor in year y
Source of data	Project calculations based on data from CAMMESA's website and IPCC 2006
Value applied	0.448
Justification of choice of data or description of measurement methods and procedures applied	Paragraph 54, Equation 9 of Tool 07.0 Version 07.0
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO2/MWh
Description	Build margin CO2 emission factor in year y
Source of data	Project calculations based on data from CAMMESA's website and IPCC 2006
Value applied	0.184
Justification of choice of data or description of measurement methods and procedures applied	Paragraph 77, Equation 15 of Tool 07.0 Version 07.0
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$W_{OM} = W_{BM}$
Data unit	-
Description	Weighting of the operating margin and the build margin emissions factor
Source of data	Tool 07.0 Version 07.0
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	Paragraph 86 option (b) of Tool 07.0 Version 07.0
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO2/MWh
Description	Combined margin emissions factor
Source of data	Project calculations based on data from CAMMESA's website and IPCC 2006
Value applied	0.316
Justification of choice of data or description of measurement methods and procedures applied	Paragraph 85, Equation 16 of Tool 07.0 Version 07.0
Purpose of Data	Calculation of baseline emissions
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	Nda,y
Data unit	Number
Description	Number of days animal is alive in the farm in the year y
Source of data	Project proponent.
Description of measurement methods and procedures to be applied	The livestock stock is actualized daily by the foreman of BiodeS which reports deaths, entries, and outputs in the moment they happen. This is reported at an excel spreadsheet where administration has access and control.
Frequency of monitoring/recording	Annually, based on monthly records.
Value applied	191
Monitoring equipment	Not applicable
QA/QC procedures to be applied	The consistency between the value and indirect information (records of sales, records of food purchases) will be assessed. The incomes and outcomes of animals in the project activity are registered in SENASA, National Service for Food Health and Quality (from the Spanish: Servicio Nacional de Sanidad y Calidad Agroalimentaria)
Purpose of data	Calculation of baseline emissions
Calculation method	Not applicable
Comments	—

Data / Parameter	$N_{p,y}$
Data unit	Number
Description	Number of animals produced annually of type LT for the year y
Source of data	Project proponent.
Description of measurement methods and procedures to be applied	The livestock stock is actualized daily by the foreman of BiodeS which reports deaths, entries, and outputs in the moment they happen. This is reported at an excel spreadsheet where administration has access and control
Frequency of monitoring/recording	Annually, based on monthly records.
Value applied	6,609
Monitoring equipment	Not applicable
QA/QC procedures to be applied	The consistency between the value and indirect information (records of sales, records of food purchases) will be assessed. The incomes and outcomes of animals in the project activity are registered in SENASA, National Service for Food Health and Quality (from the Spanish: Servicio Nacional de Sanidad y Calidad Agroalimentaria)
Purpose of data	Calculation of baseline emissions
Calculation method	Not applicable
Comments	—

Data / Parameter	Wsite																
Data unit	Kg																
Description	Average animal weight of a defined livestock population at the project site																
Source of data	Project proponent.																
Description of measurement methods and procedures to be applied	A weigh bridge scale at the entrance/exit of BiodeS installations where livestock is weighted at the entry and when they leave. This is the official balance, approved by the INTI (National Institute of Industrial Technology). Another electronic scale is installed in the animal chutes, where they make individual controls to analyse the behaviour of the lot.																
Frequency of monitoring/recording	Annually																
Value applied	300																
Monitoring equipment	<table border="1"> <tr> <td>Data</td><td>Scale</td></tr> <tr> <td>Manufacturer</td><td>LATORRE</td></tr> <tr> <td>Model</td><td>BSB</td></tr> <tr> <td>Size</td><td>25 x 3 m</td></tr> <tr> <td>Accuracy class</td><td>III</td></tr> <tr> <td>Mechanic gauge, model</td><td>L-83</td></tr> <tr> <td>Electronic gauge, model</td><td>LE 100 1</td></tr> <tr> <td>Calibration frequency</td><td>Yearly</td></tr> </table>	Data	Scale	Manufacturer	LATORRE	Model	BSB	Size	25 x 3 m	Accuracy class	III	Mechanic gauge, model	L-83	Electronic gauge, model	LE 100 1	Calibration frequency	Yearly
Data	Scale																
Manufacturer	LATORRE																
Model	BSB																
Size	25 x 3 m																
Accuracy class	III																
Mechanic gauge, model	L-83																
Electronic gauge, model	LE 100 1																
Calibration frequency	Yearly																
QA/QC procedures to be applied	The monitoring equipment will be calibrated periodically according with manufacturer's recommendations or local/international standard to ensure accuracy.																
Purpose of data	Calculation of baseline emissions																
Calculation method	As per the "Standard for sampling and surveys for CDM project activities and Programmes of Activities"																

Comments	—
Data / Parameter	VSLT,y
Data unit	kg dry matter/animal/year
Description	Volatile solids for livestock LT entering the animal manure management system in year y
Source of data	Laboratory analysis
Description of measurement methods and procedures to be applied	The laboratory protocols of analysis will be available at the validation visit
Frequency of monitoring/recording	Annually
Value applied	1.998
Monitoring equipment	Not applicable
QA/QC procedures to be applied	The amount of VS excreted is consistent with the diet and the animal weight
Purpose of data	Calculation of baseline emissions
Calculation method	Not applicable
Comments	—

Data / Parameter	$BG_{burnt,y}$
Data unit	Nm ³ /y
Description	Biogas volume in year y
Source of data	On site measurement.
Description of measurement methods and procedures to be applied	The volumetric flow is measured by flow meters and data is shown in SCADA and recorded by SCADA operator daily.
Frequency of monitoring/recording	Annually, based on continuous flow measurement with accumulated volume recording.
Value applied	2,163,936
Monitoring equipment	Flow meter
QA/QC procedures to be applied	Calibration and frequency of calibration of the flow meter is according to manufacturer's specifications
Purpose of data	Calculation of baseline emissions
Calculation method	Not applicable
Comments	—

Data / Parameter	W_{CH_4}
Data unit	%
Description	Methane content in biogas in the year y
Source of data	On site measurement.
Description of measurement methods and procedures to be applied	The methane content in the biogas is analysed with a continuous analyser
Frequency of monitoring/recording	Every 30 minutes.
Value applied	54.8%
Monitoring equipment	Continuous analyser
QA/QC procedures to be applied	The electronic gas analyser will be calibrated according to technology provider recommendations.
Purpose of data	Calculation of baseline emissions
Calculation method	An average of the data accumulated in the year y.
Comments	—

Data / Parameter	T
Data unit	°C
Description	Temperature of the biogas at the flow measurement site
Source of data	Biogas flow meter measurements. No separate monitoring of temperature is required when using flow meters that automatically measure temperature and pressure.
Description of measurement methods and procedures to be applied	The biogas flow meter measures flow, pressure and temperature and displays or outputs the normalised flow of biogas.
Frequency of monitoring/recording	Continuously together with the flow measurements.
Value applied	Monitored ex post by the biogas flow meter
Monitoring equipment	The biogas flow meter measures flow, pressure and temperature and displays or outputs the normalised flow of biogas.
QA/QC procedures to be applied	The biogas flow meter will be maintained and calibrated periodically according with manufacturer's recommendations or local/international standard to ensure accuracy.
Purpose of data	Calculation of baseline emissions
Calculation method	Not applicable
Comments	No separate monitoring of temperature is necessary when using flow meters that automatically measure temperature and pressure, expressing biogas flow in normalized cubic meters.

Data / Parameter	P
Data unit	Pa
Description	Pressure of the biogas at the flow measurement site
Source of data	Biogas flow meter measurements. No separate monitoring of temperature is required when using flow meters that automatically measure temperature and pressure.
Description of measurement methods and procedures to be applied	The biogas flow meter measures flow, pressure and temperature and displays or outputs the normalised flow of biogas.
Frequency of monitoring/recording	Continuously together with the flow measurements.
Value applied	Monitored ex post by the biogas flow meter
Monitoring equipment	The biogas flow meter measures flow, pressure and temperature and displays or outputs the normalised flow of biogas.
QA/QC procedures to be applied	The biogas flow meter will be maintained and calibrated periodically according with manufacturer's recommendations or local/international standard to ensure accuracy.
Purpose of data	Calculation of baseline emissions
Calculation method	Not applicable.
Comments	No separate monitoring of temperature is necessary when using flow meters that automatically measure temperature and pressure, expressing biogas flow in normalized cubic meters.

Data / Parameter	$Q_{(stored,y)}$
Data unit	m ³ /y
Description	Amount of liquid digestate stored anaerobically in year y
Source of data	Plant records.
Description of measurement methods and procedures to be applied	The measurement is made by level of the secondary tank, there is an internal level sensor that measures the emptying height. With the height and the area of the tank, the emptied volume is calculated
Frequency of monitoring/recording	Record daily, annually calculated
Value applied	36,500
Monitoring equipment	Level sensor
QA/QC procedures to be applied	Not applicable
Purpose of data	Calculation of project emissions.
Calculation method	Level of liquid eliminated x area of the biodigester= volume of digestate
Comments	The plant operators record daily the level of liquid eliminated from the biodigesters. The equation establishes the volume in m ³

Data / Parameter	$P_{\text{COD},y}$
Data unit	t COD / m ³
Description	Average chemical oxygen demand (COD) of the liquid digestate in year y
Source of data	Laboratory analysis
Description of measurement methods and procedures to be applied	The laboratory protocols of analysis will be available at the validation visit.
Frequency of monitoring/recording	Monthly and averaged annually.
Value applied	0.02182
Monitoring equipment	Not applicable.
QA/QC procedures to be applied	The quality assurance and quality control (QA/QC) procedures to be applied, are those applied in the laboratory analysis of the digestate.
Purpose of data	Calculation of project emissions.
Calculation method	Not applicable
Comments	The project developer will irrigate the liquid digestate in some periods of the year, therefore the average COD during these periods will be 0. The leakage emissions will be registered in the cases where the liquid outflow is stored in the open lagoon for more than 1 month.

Data / Parameter	nd _y
Data unit	Days
Description	Number of days that the animal manure management system was operational.
Source of data	Plant records.
Description of measurement methods and procedures to be applied	Not applicable
Frequency of monitoring/recording	Annually, based on daily records and monthly aggregation.
Value applied	365
Monitoring equipment	Not applicable
QA/QC procedures to be applied	Not applicable
Purpose of data	Calculation of baseline emissions
Calculation method	Not applicable
Comments	-

Data / Parameter	$EG_{(PJ, facility, y)}$
Data unit	MWh/y
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Electricity meter
Description of measurement methods and procedures to be applied	This parameter will be monitored using bi-directional energy meter Class 0.5S
Frequency of monitoring/recording	Continuously measured, monthly recorded
Value applied	4,641
Monitoring equipment	Electricity meter class 0.5S
QA/QC procedures to be applied	Electricity meter will be calibrated according to CAMMESA procedures. In case of failures CAMMESA procedure will be followed. The electricity generation will be cross-checked with invoices.
Purpose of data	Calculation of baseline emissions
Calculation method	Not applicable
Comments	—

Data / Parameter	B_0
Data unit	tCH ₄ /tCOD
Description	Maximum methane producing potential of the COD applied
Source of data	Default value in table 6.2 of the IPCC 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 5 Chapter 6.
Description of measurement methods and procedures to be applied	Measured through laboratory analysis.
Frequency of monitoring/recording	Every time the liquid digestate is not irrigated for more than one month
Value applied	0.25
Monitoring equipment	Not applicable
QA/QC procedures to be applied	Soil applications will be registered and controlled by the project proponent. This information will be crosschecked with the dates of analysis of the digestate.
Purpose of data	To calculate Leakage emissions
Calculation method	Not applicable
Comments	—

Data / Parameter	Soil application of the outflow
Data unit	Not applicable
Description	Soil application of the solid and liquid digestate
Source of data	Project proponent records.
Description of measurement methods and procedures to be applied	The application of the liquid and solid digestate will be monitored and registered to verify that is a positive practice for the composition of the soil. Periodic soil analysis will be carried out to ensure the correct application in time and quantity. The solid fraction is going to be applicated as biofertilizer and the liquid fraction of digestate is going to be irrigated every day. The emissions generated during the lapse time between the stages of the crops, where the liquid cannot be applicated, are considered as project emissions.
Frequency of monitoring/recording	Every time the digestate is applicated or irrigated.
Value applied	Not applicable
Monitoring equipment	Not applicable
QA/QC procedures to be applied	Soil applications will be registered and controlled by the project proponent.
Purpose of data	To demonstrate that there are no project emissions related to the effluents when they are applicated.
Calculation method	Not applicable
Comments	—

5.3 Monitoring Plan

The emission reductions achieved by the project activity will be determined ex post through direct measurement of the amount of methane flared. It is likely that the project activity involves manure treatment steps with higher methane conversion factors (MCF) than the MCF for the manure treatment systems used in the baseline situation, therefore the emission reductions achieved by the project activity are limited to the ex post calculated baseline emissions minus the project emissions using the actual monitored data for the project activity (i.e. $N_{LT,y}$, $MS\%_{i,y}$, as well as $VS_{LT,y}$ in cases where adjusted values for animal weight are used). The emission reductions achieved in any year are the lowest value of the following:

$$ER_{y,ex\ post} = \min [(BE_{y,ex\ post} - PE_{y,ex\ post}) , (MD_y - PE_{power,y,ex\ post})] \quad \text{Equation (10)}$$

Where:

- $ER_{y,ex\ post}$ = Emission reductions achieved by the project activity based on monitored values for year y (t CO₂e)
- $BE_{y,ex\ post}$ = Baseline emissions calculated using equation 1 (for projects using option in paragraph 17(a)) using ex post monitored values of $N_{LT,y}$ and if applicable $VS_{LT,y}$. For projects using option in paragraph 17(b), the ex post monitored values for $Q_{manure,j,LT,y}$ and $SVS_{j,LT,y}$ are used
- $PE_{y,ex\ post}$ = Project emissions calculated using equation 6 using ex post monitored values of $N_{LT,y}$, $MS\%_{i,y}$, $MS\%_I$, AI_I , $Q_{res\ waste,y}$ and if applicable $VS_{LT,y}$
- MD_y = Methane captured and destroyed or used gainfully by the project activity in year y (t CO₂e)
- $PE_{power,y,ex\ post}$ = Emissions from the use of fossil fuel or electricity for the operation of the installed facilities based on monitored values in the year y (t CO₂e)

As the project activities utilize the recovered methane for power generation, MD_y may be calculated as follows, based on the amount of monitored electricity generation, without monitoring methane flow and concentration:

$$\underline{EG_y \times 3600} \quad \text{Equation (12)}$$

$$MD_y = NCV_{CH4} \times EE_y \times D_{CH4} \times GWP_{CH4}$$

Where:

- EG_y = Total electricity generated from the recovered biogas in year y (MWh)
- 3600 = Conversion factor (1 MWh = 3600 MJ)

NCV_{CH_4} = NCV of methane (MJ/Nm³) (use default value: 35.9 MJ/Nm³)

EE_{yy} = Energy conversion efficiency of the project equipment, which is determined by adopting one of the following criteria:

- Specification provided by the equipment manufacturer. The equipment shall be designed to utilize biogas as fuel, and efficiency specification is for this fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation.
- Default efficiency of 40 %.

Methods for measuring, recording, storing, aggregating, collating and reporting data and parameters.

The plant is operated and continuously controlled and monitored by a SCADA system supplied by BGA. This allows remote monitoring of all the plant.

This software controls, among others, the biomass loaded to the biodigesters, the biogas production and the energy generation. Controls quantities and proportions of biomass, as well as quality and quantity of biogas produced. The system includes alarm management of minimum and maximum levels. All the monitored data is registered automatically in the system.

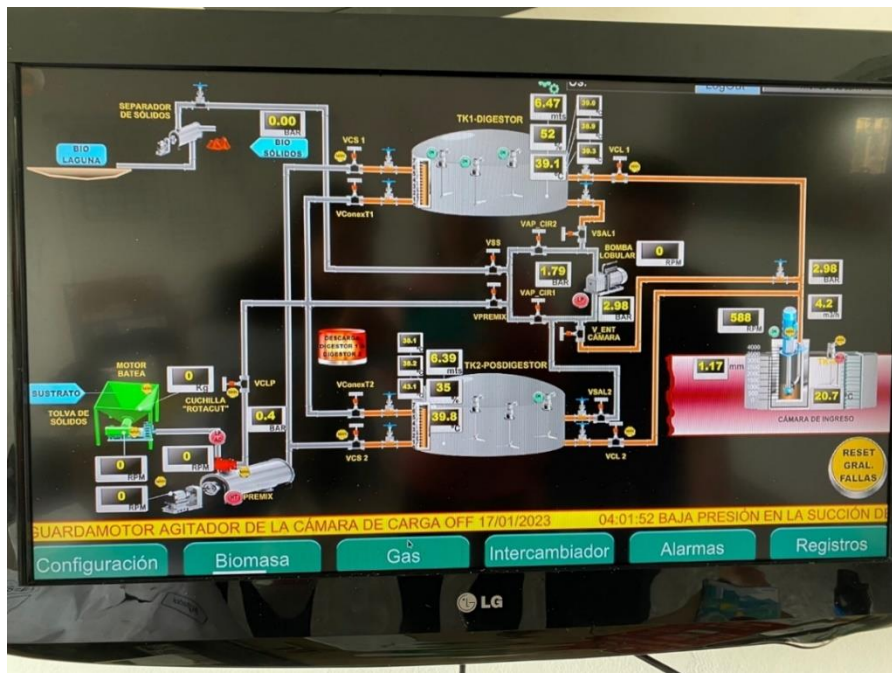


Figure 10. SCADA system

The electricity generated is delivered to the national grid (as stated before), for this reason the source of data to be used of the electricity supplied by the engine generator is the master energy meter that belongs to the Commercial Measurement System ("SMEC" from

the Spanish Sistema de Medición Comercial) (Official data). Measuring, recording, storing, aggregating, collating, and reporting data and parameters must follow the Wholesale Electricity Market Administrator Company (CAMMESA) technical procedures regarding the Commercial Measurement System and the Real Time Operation System (“SOTR” from the Spanish Sistema de Operación en Tiempo Real).

Organizational structure, responsibilities, and competencies of the monitoring personnel

The maintenance supervisors of BGA are responsible for controlling the plant and collecting data monitored on-site. They are technicians with vast experience in the equipment operation, maintenance and calibration, and emergency procedures.

BiodeS personnel were trained with special attention in operation of the Scada system and on-site laboratory analysis to ensure that the critical parameters are continuously controlled.

Scada system alerts (alarm signals) about failures. Clear procedures to be followed by operators and personnel are established for emergency situations.

Overall responsibility for the monitoring and maintenance of all required tasks and their adequate management lies with the project manager.

Quality Assurance and Quality control

BiodeS performs the internal controls with the SCADA system provided by BGA. The monitoring data is filed electronically, and all the files are backed-up.

A detailed monitoring plan is available for the operators in the plant to ensure the right functioning of all the equipment and the control of all the critical parameters.

The Monitoring Plan will be implemented over the 7- year crediting periods of project activity. All data and evidence collected as part of monitoring will be archived electronically and be kept at least for 2 years after the end of the last crediting period. All printed documentation is stored and available to be checked in the validation/verification audits.

Calibration of the equipment:

With regards to the biogas generation, maintenance supervisors of BGA calibrated and control all the meters and points of control in the plant.

In respect to the electricity generation, the energy meters are property of SMEC, SMEC is therefore responsible for their maintenance. The meters are sealed to avoid unauthorized interventions. (CAMMESA is responsible for the inspection of the meters). The maintenance will be conducted in accordance with national procedures and standards.